CPTC31081 · ESSEC MIM 1 · FALL 2026

# Refresher Course Accounting

Full revision notes, Class #1 to Class #10

> Final exam on Tuesday 15 December 2026 in the morning (DOME), on paper: 100% of the grade - financial accounting 60%, management accounting 40%. Four-function calculator only. Resit early 2027.
> Professor: Sameh Amer.
> Sources: FA 2026-2027 slides (Class #1-#6), MA slides (Class #7-#10), Workbook, Moodle exercises and sample exams.

**How to use these notes.** Each class gives the key concepts, the mechanisms step by step, a worked example and the traps. The practice section of each class has every exercise with its statement and a step-by-step correction (official when published on Moodle, clearly marked "Proposed solution" otherwise). The MCQ annex covers all classes with answers and explanations.

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# Class #1 Overview of financial accounting (9 Oct.)

Readings: Stolowy et al., ch. 1 to 5 (pages in the course outline). Workbook cases: Verdi, Busoni.

**In one look.**

- = Assets = Liabilities + Shareholders' equity
- = Net income = Revenue − Expenses (included in equity)
- Every transaction hits both sides by the same amount, or reclassifies within one side.
- Revenue or expense only if equity moves, excluding contributions and dividends.
- Net income ≠ cash (borrowing, equipment, credit sales).
- Statements: balance sheet, income statement, cash flow statement, statement of changes in equity, notes.
- Historical cost (verifiable) vs fair value (market); IFRS 18 replaces IAS 1 in 2027.

## Why financial accounting matters

- Accounting is the "language of business". Financial accounting reports the resources a company controls and how management uses them.
- Users: shareholders, lenders, suppliers, customers, employees, analysts, the State, the public, management. Information that meets investors' needs is assumed to meet everyone's.
- This course takes the **user's** perspective: read and interpret financial statements.

## Key concepts · The business equation

```
Assets = Liabilities + Shareholders' equity
Net income = Revenue - Expenses   (net income stays inside equity)
```

| Term | Course definition (IASB 2018 conceptual framework) |
|---|---|
| **Asset** | A present economic resource controlled by the entity as a result of past events; an economic resource is a right that has the potential to produce economic benefits. Current or non-current. |
| **Liability** | A present obligation to transfer an economic resource as a result of past events. Current or non-current. |
| **Shareholders' equity** | The residual interest in the assets after deducting liabilities: share capital + retained earnings. |
| **Revenue** | An increase in assets or decrease in liabilities that increases equity, excluding shareholders' contributions. |
| **Expense** | A decrease in assets or increase in liabilities that decreases equity, excluding distributions to shareholders. |

## Mechanism · How every transaction keeps the equation balanced

1. Identify what the company receives and what it gives up or owes.
2. Every transaction either changes both sides by the same amount (+X / +X or -X / -X) or reclassifies within one side (+X and -X).

3. Then ask: **does it touch shareholders' equity?** If yes, it is revenue or an expense - unless it is a shareholders' contribution or a dividend.

**Net income is not cash.** Some transactions move cash without touching income (borrowing, buying equipment, repaying a loan); others hit income and cash in different periods (credit sales, credit purchases). That is the accruals principle.

## Worked example · Case Verdi (Workbook, official)

**Verdi · the business equation transaction by transaction** Class #1 slides + Workbook **Official solution**

Stefania and Stefano create the advertising agency Verdi. Track each transaction in the equation (CU):

| # | Cash | Equipment | Receivables | = | Payables | Bank loans | + | Share capital | Net income | Detail |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | +150 | | | | | | | +150 | | Owners invest 150 |
| 2 | +60 | | | | | +60 | | | | Bank loan |
| 3 | -125 | +125 | | | | | | | | Equipment bought |
| 4 | | | +250 | | | | | | +250 | Services rendered (revenue) |
| 5 | +180 | | -180 | | | | | | | Client pays part of invoice |
| 6a | -101 | | | | | | | | -101 | Salaries expense |
| 6b | -4 | | | | | | | | -4 | Interest expense |
| 6c | | | | | +85 | | | | -85 | Subcontractor invoice |
| 7 | -80 | | | | -80 | | | | | Partial payment to supplier |
| 8 | -12 | | | | | -12 | | | | Partial loan repayment |
| **End** | **68** | **125** | **70** | | **5** | **48** | | **150** | **60** | 263 = 53 + 210 |

▼ **Verdi's three financial statements (official)**

**Balance sheet, 31 Dec X1** - Equipment 125; accounts receivable 70; cash 68. Total assets **263** = share capital 150 + net income 60 + bank loans 48 + accounts payable 5.

**Income statement, year X1** - Revenue 250 (sales) - expenses 190 (third-party 85, personnel 101, interest 4) = net income **60**.

**Statement of cash flows, year X1** - Operating: +180 clients, -80 suppliers, -101 employees = **-1**. Investing: -125 equipment. Financing: +150 shares, +60 loan, -12 repayment, -4 interest = **+194**. Net change +68: closing cash **68**.

## The financial statements

- **Statement of financial position (balance sheet)**: uses of resources (assets) and their origins (liabilities and equity) at a point in time.
- **Income statement** (statement of profit or loss): revenue - expenses over the period.
- **Statement of changes in equity**, **statement of cash flows** (operating, investing, financing) and **notes**.
- The statements are linked: net income feeds equity; the change in cash explains the cash line of the balance sheet.

## The rules of the game

- Why rules: to value assets and liabilities, recognize revenue and expenses, and separate an asset from an expense (a share bought for 5 worth 30 at year-end; a pool table sold on 31/12, delivered on 31/01; 60M of R&D).
- Standard setters: the IASB (IAS/IFRS) and the ISSB (sustainability, IFRS S1 and S2, since 2021) within the IFRS Foundation; the EU ("IFRS as adopted by the EU"); the FASB (US GAAP); the ANC (French rules). US GAAP is more rules-based, IFRS more principles-based. The 1993 Daimler-Benz example: a profit of 615M DM under German rules, a loss of 1,839M DM under US GAAP.
- Principles: going concern, faithful representation (complete, neutral, free from error), consistency, accruals, prudence, monetary unit.
- Measurement bases: **historical cost** (easy, verifiable, but blind to market value) and **current value**, including **fair value** (market price; mostly financial instruments; IAS 16 allows revaluation, IAS 40 fair value for investment property).
- Governance: shareholders elect the board, which appoints management and an audit committee; the auditor certifies the accounts.

**2027 change (IFRS 18).** IFRS 18 replaces IAS 1 for periods starting on or after 1 January 2027. The income statement is split into five categories (operating, investing, financing, tax, discontinued operations) with two subtotals: operating profit and profit before financing and tax. This course presents the 2026 statements.

## Classic traps

- Collecting a receivable or paying a payable does not touch income.
- A loan is not revenue; repaying it is not an expense; only the interest is an expense.
- A shareholders' contribution increases equity without being revenue.

# Class #2 Operating cycle and current assets (16 Oct.)

Readings: Stolowy et al., ch. 6, 9 and 10. Cases: Mozart (Workbook), Schultz exercise.

**In one look.**

- IFRS 15: contract → obligations → price → allocation → revenue when the obligation is satisfied.
- Revenue: goods at a point in time (transfer of control), services over time.
- = Acquisition cost = price + transport + duties − discounts
- = Cost of goods sold = opening inventory + purchases − closing inventory
- Rising prices: FIFO → higher income and inventory; LIFO → the opposite (banned under IFRS).
- Inventory at the lower of cost and net realisable value.
- Doubtful receivable → impairment (expense); uncollectible → written off.

## Key concepts · The operating cycle

Purchases → inventories → production → sales → receivables → collections; purchases → payables → payments. **Current assets** are used within one cycle and turn into cash within a year: inventories, trade receivables, tax receivables, advances to suppliers, cash.

## Mechanism · Recognizing revenue and expenses

| | Accrual accounting | Cash accounting |
|---|---|---|
| Revenue | When the performance obligation is satisfied (at a point in time or over time) | When collected |
| Expenses | When incurred | When paid |

1. **IFRS 15 five-step model**: identify the contract; identify the performance obligations; determine the transaction price; allocate it to the obligations; recognize revenue when (or as) each obligation is satisfied.
2. Goods: at a point in time (transfer of control); services: over time. LVMH example: retail sales recognized when the customer buys, services as rendered, net of discounts.
3. The ideal is to **match expenses to the revenue** they helped produce.

**Revenue recognition is the fraud hotspot.** In 60% of the US scandals studied by COSO, revenue recognition was manipulated: period shifts ("cookie jar") or fictitious sales (Gowex, Satyam). See also Tesco (2014).

## Inventories

- Until a product is sold, its costs stay on the balance sheet. Measured at the **lower of cost and net realizable value**.
- Types: merchandise, raw materials, finished goods.

```
Acquisition cost = purchase price + costs to bring the inventory on site and to condition
```

```
                    (transport, handling) + directly attributable costs (customs,
non-recoverable taxes) - discounts
Production cost = direct costs + a systematic share of production overheads
Cost of goods sold = opening inventory + purchases - closing inventory
```

| Method | Assumption | Effect |
|---|---|---|
| Specific identification | Non-fungible goods | Actual cost of each item |
| FIFO | First in, first out | Closing inventory near recent prices (balance-sheet oriented) |
| LIFO | Last in, first out | Banned under IFRS, allowed under US GAAP; reacts fast to prices (income-statement oriented) |
| WAC | Weighted average cost, recomputed at each purchase | No assumed order |

## Worked example · Case Cameras (official)

**Cameras · WAC vs FIFO vs LIFO** Class #2 slides **Official solution**

Opening inventory 1 Jan: 50 cameras at 100 euros. 10/01: buy 75 at 110. 15/01: sell 60 at 150. 20/01: buy 25 at 120. 30/01: sell 40 at 150. Closing inventory and impact under each method?

▼ **See the official solution**

**WAC**: on 15/01 the average is (50×100 + 75×110)/125 = 106 → COGS 60×106 = 6,360, inventory 65×106 = 6,890. On 30/01: (65×106 + 25×120)/90 = 109.9 → COGS 40×109.9 = 4,396; closing inventory 50×109.9 = **5,495**.

**FIFO**: on 15/01 out go 50×100 + 10×110 = 6,100; on 30/01 out go 40×110 = 4,400; closing inventory 25×110 + 25×120 = **5,750**.

**LIFO**: on 15/01 out go 60×110 = 6,600; on 30/01 out go 25×120 + 15×110 = 4,650; closing inventory 50×100 = **5,000**.

| | FIFO | WAC | LIFO |
|---|---|---|---|
| Sales | 15,000 | 15,000 | 15,000 |
| COGS | 10,500 | 10,756 | 11,250 |
| Profit | 4,500 | 4,244 | 3,750 |
| Closing inventory | 5,750 | 5,495 | 5,000 |

When prices **rise**, FIFO maximizes profit and closing inventory and minimizes COGS; LIFO does the opposite. When prices fall, the effects reverse.

## Accounts receivable

- They exist because transactions are not settled instantly and accounting is accrual-based.

- **Doubtful receivable**: an allowance on the balance sheet, with an expense in the income statement. **Uncollectible receivable**: written off, with a loss. Under IFRS 9 (simplified approach), trade receivables are impaired from day one for lifetime expected credit losses.

# Class #3 Investing cycle and non-current assets (23 Oct.)

Readings: Stolowy et al., ch. 7 and 8. Case: Amorti (Workbook).

**In one look.**

- Non-current = used over more than one cycle (purpose, not nature).
- Entry cost = price + directly attributable costs (100 + 5 + 15 + 30 = 150).
- = Straight-line = (cost − residual value) / useful life
- = Double declining balance = carrying amount × 2 / useful life
- = Gain on disposal = sale proceeds − carrying amount
- Land, brands and goodwill: no depreciation. Advertising and research: expensed.

## Key concepts · Current vs non-current

- A **non-current asset** is not meant to be turned into cash within the normal cycle (usually one year). The difference with a current asset is **use**: the same car is a fixed asset for one company and inventory for a car dealer.

| Type | Examples | Consumption |
|---|---|---|
| Property, plant & equipment | Buildings, equipment, machines | Depreciation |
| Land | | No depreciation |
| Natural resources | Deposits, quarries | Depletion |
| Finite-life intangibles | Patents, software, licences, franchises | Amortization |
| Indefinite-life intangibles | Brands, goodwill | No amortization (impairment tests) |
| Financial assets | Securities, long-term loans | No amortization; amortized cost or fair value (IFRS 9) |

## Mechanism · Initial recognition at cost

1. Start from the acquisition cost (if bought) or production cost (if built).
2. Add the directly attributable costs needed to bring the asset to working condition: site preparation, delivery, installation, architects' and engineers' fees, estimated dismantling costs.
3. Slide example: price 100 + customs 5 + transport 15 + fees 30 = equipment at **150**.

- **Intangibles**: identifiable non-monetary assets without physical substance, recognized if future benefits are probable and the cost is measurable (e.g. development costs). No asset for internally generated brands, internal goodwill, customer lists, advertising, training or research. Only **acquired goodwill** is recognized (price paid - fair value of net identifiable assets; LVMH and Tiffany in 2021).

## Mechanism · Depreciation

```
Depreciable amount = cost - residual value
Straight-line: rate = 1 / useful life; annual charge = depreciable amount x rate
Declining balance (DDBM): rate = 2 x straight-line rate; charge = net book value x rate
Net book value = cost - accumulated depreciation (- impairment)
Gain or loss on disposal = sale price - net book value
```

- Partial year: pro-rate the charge (months or days).
- Accelerated methods charge more in early years (cars, ovens) and bring a tax shield. In the US the multiple is often 2; in continental Europe the tax authorities set coefficients by useful life.

### Worked example · Straight-line and declining balance (official)

**Two depreciation schedules** Class #3 slides **Official solution**

(1) Cost 11,000, residual value 1,000, 5 years, straight-line. (2) NCA machine of 15,000 over 10 years, double declining balance: charges for the first two years.

▼ See the official solution

| Year-end | Charge | Cumulated | NBV |
|---|---|---|---|
| X1 | 2,000 | 2,000 | 9,000 |
| X2 | 2,000 | 4,000 | 7,000 |
| X3 | 2,000 | 6,000 | 5,000 |
| X4 | 2,000 | 8,000 | 3,000 |
| X5 | 2,000 | 10,000 | 1,000 |

(1) Depreciable amount 10,000 × 20% = 2,000 per year. (2) Straight-line rate 10% → DDBM 20%. Year 1: 15,000 × 20% = **3,000** (NBV 12,000). Year 2: 12,000 × 20% = **2,400**; cumulated 5,400; NBV 9,600.

### Derecognition

On disposal, or when no future benefit is expected: remove the net book value and recognize the net result of the sale (IAS 16.68). Slide example: equipment with an NBV of 40M sold for 45M → gain of 5M.

Depreciation is an expense with **no cash effect** (except through tax). Book value is usually **not** market value: hidden gains can exist. Always challenge the assumptions (useful life, pattern of consumption).

# Class #4 Financing cycle, equity and liabilities (30 Oct.)

Readings: Stolowy et al., ch. 11 and 12. Cases: ACME (Workbook), Gerry Co. exercise.

**In one look.**

- = Share capital = shares × par value
- = Share premium = (issue price − par value) × shares issued
- = Book value per share = equity / shares outstanding
- Appropriation: reserves, dividends (a liability once declared but unpaid), retained earnings.
- Debt: repayment, accrued and deductible interest; equity: no repayment, control.

## Key concepts · Equity vocabulary

```
Share capital = number of shares x par value
Book value per share = shareholders' equity / number of shares
Market capitalization = number of shares x share price
Share premium = (issue price - par value) x new shares issued
```

- **Par value** only defines the number of shares in the legal capital and each shareholder's power. Par ≠ book value ≠ market value.
- The **share premium** (additional paid-in capital, capital surplus) records what new shareholders pay above the legal capital; it belongs to equity.

## Mechanism · Issuing new shares

1. New shares x par value → increases **share capital**.
2. New shares x (issue price - par value) → increases **share premium**.
3. New shares x issue price → cash coming in.

## Profit appropriation

- Three destinations: **reserves** (legal, statutory, free), **dividends**, **retained earnings**.
- Three cases: keep everything in reserves, distribute everything, or a mix. Dividends decided but not yet paid become a **liability** (dividends payable).
- The balance sheet is presented **before** appropriation (the year's income on its own line) or **after** (reserves updated, dividends in liabilities).

## Liabilities

A liability is a present obligation to transfer an economic resource as a result of a past event (legally enforceable or customarily accepted; settled in cash, goods, services, shares…).

## Debt vs equity

| | Debt | Equity |
|---|---|---|
| Repayment | Mandatory | None |

| | | |
|---|---|---|
| Flexibility | High (term, amount, refinancing) | Low, high transaction costs, irreversible decision |
| Cost | Interest due whatever the income, market rate | No mandatory return; dividends if profits; higher expected return (risk) |
| Tax | Interest usually deductible | Dividends usually not deductible |
| Power | No control over management | Influence and control; dilution with new shareholders |
| Use of funds | Can be tied to a project (guarantees, mortgages) | Not earmarked |

Debt and equity play different risk/reward roles. Some companies have only private equity yet issue public debt (bonds). Analysts, shareholders and creditors all read this split.

# Class #5 End-of-Year adjustments (6 Nov.)

Readings: Stolowy et al., ch. 4, 7 and 12. Cases: Flossie Flowers, Imp, Provisor (Workbook), Offenbach exercise.

**In one look.**

- Prepaid expense (asset), deferred revenue (liability), accrued expense (liability), accrued revenue (asset).
- = Recoverable amount = max(fair value − costs of disposal; value in use)
- = Impairment loss = carrying amount − recoverable amount
- Provision: present obligation + probable outflow + reliable estimate; otherwise a contingent liability in the notes.
- No provision for future losses; restructuring only once a formal plan is announced.

## Why year-end adjustments

Depreciation and impairment, accruals and deferrals, provisions, doubtful receivables: they apply the accruals, matching and prudence principles, cutting activity into annual periods at the closing date (often 31 December). The slides show a company can be very profitable with dangerously low cash (clients paying at two months, suppliers at one), or loss-making with rising cash.

## Mechanism · Accruals and deferrals

| Situation | Cash | Expense / revenue | Closing entry |
|---|---|---|---|
| **Prepaid expense** | Paid in N | Relates to N+1 | Asset "prepaid expense"; reduce the expense of N |
| **Deferred revenue** | Collected in N | Relates to N+1 | Liability "deferred revenue"; reduce the revenue of N |
| **Accrued expense** | Paid in N+1 | Relates to N | Liability "accrued expense"; expense of N |
| **Accrued revenue** | Collected in N+1 | Relates to N | Asset "accrued revenue"; revenue of N |

| Slide example | Entry at 31/12/N |
|---|---|
| Rent of 9,000 paid on 01/11/N for Nov, Dec and Jan | Prepaid expense +3,000; expense of N = 6,000 |
| December electricity bill of 100, received and paid on 05/01/N+1 | Accrued expense +100; expense of N = 100 |
| Annual subscription of 1,200 collected on 01/07/N, all in revenue | Deferred revenue +600; revenue of N = 600 |
| Invoice of 900 issued on 01/02/N+1 for Nov, Dec and Jan | Accrued revenue +600; revenue of N = 600 |

## Mechanism · Impairment of assets

```
Recoverable amount = max(fair value - costs of disposal; value in use)
Impairment loss = carrying amount - recoverable amount   (if positive)
```

- Prudence: an asset may not sit on the books above its recoverable amount. Value in use = present value of the asset's expected cash flows.
- One-off test, usually at closing. Reversal allowed if the recoverable amount rises again (IAS 36.114), capped at what the carrying amount would have been without impairment; **never for goodwill**.

## Provisions

| | Cause | Timing | Amount |
|---|---|---|---|
| Certain liability | Real and present | Certain | Certain |
| Accrued expense | Real and present | Near certain | Uncertain |
| **Provision** | Real and present | Uncertain | Uncertain |
| Contingent liability | Possible obligation, or present but unrecognized | Uncertain | Uncertain |

- A provision is a liability of uncertain timing or amount. Three **cumulative** conditions: a present obligation (legal or constructive) from a past event; a probable outflow of resources; a reliable estimate. Otherwise no provision, only a **contingent liability** disclosed in the notes.
- Accounting: an expense and a provision in liabilities, at the best estimate. Later: use (liability and cash both fall) or reversal (liability falls, revenue is recognized).

| Slide case | Provision? |
|---|---|
| Shop that refunds unhappy customers under a well-known policy, no legal obligation | Yes: constructive obligation, if outflow probable and estimable |
| Oil company polluting in a country with no law; clean-up law near certain just after the 20X0 closing | No in earlier years; yes in 20X0 because of the new law |
| Division closure decided on 12/12, not announced nor started | No: the obligation arises from the announcement |
| Warranty: 75% no defect, 20% small defect (1), 5% large defect (10), 10,000 products | Yes: 10,000×20%×1 + 10,000×5%×10 = **7,000** |
| Food poisoning after a wedding: loss improbable in 20X0, probable in 20X1 | 20X0: contingent liability; 20X1: provision |

**Forbidden:** provisions for future operating losses. Restructuring: only with a detailed formal plan, already started or announced to those affected. Accruals are meant to cut off annual periods, so their cash implications are (very) different, and they create opportunities for earnings manipulation that analysts watch closely.

# Class #6 Statement of cash flows (13 Nov.)

Readings: Stolowy et al., ch. 16. Case: Beethoven (Workbook).

**In one look.**

- = Change in cash = operating + investing + financing cash flows
- Direct: receipts − payments; indirect: adjusted net income.
- = Indirect: net income + depreciation − Δ receivables and inventory + Δ operating payables
- Until 2026: choice for interest and dividends; from 2027: received → investing, paid → financing.

## Key concept · What the statement explains

The statement of cash flows explains the change in **cash and cash equivalents** (very liquid short-term investments with negligible risk of value change) over the period.

```
Operating cash flows (I)
+ Investing cash flows (II) = investment inflows - outflows
+ Financing cash flows (III) = financing inflows - outflows
= Net change in cash (IV)
Closing cash = opening cash + IV
```

## The three activities

| Activity | Inflows | Outflows |
|---|---|---|
| **Operating** | Sales of goods and services, royalties, commissions | Purchases, wages and social charges, taxes |
| **Investing** | Disposals of fixed assets, repayments of loans granted | Purchases of fixed assets, loans granted, securities (not cash equivalents) |
| **Financing** | Share issues, borrowings | Share buybacks, loan repayments |

**Interest and dividends.** Until 2026 (IAS 7), a choice: interest and dividends received in operating or investing; paid in operating or financing; tax in operating. From 1 January 2027 (IAS 7 amended by IFRS 18), for most entities: received in investing, paid in financing, tax in operating. In the exam, read the classification instructions.

## Mechanism · Direct and indirect methods

- **Direct**: collections from customers minus payments to suppliers, employees, etc. (from actual cash movements).
- **Indirect**: start from net income and correct it. A semi-direct variant corrects sales by the change in receivables, etc. Both methods give the same figure and concern only operating flows.
- From 2027, the indirect method starts from operating profit.

```
Indirect method: Net income
  + depreciation, amortization and impairment charges, - reversals (non-cash items)
  - gains / + losses on disposals (remove disposal results)
  - increase in inventories and receivables (+ if decrease)
  + increase in trade and operating payables (- if decrease)
= Operating cash flow
```

## Worked example · Verdi's statement of cash flows (official)

**Verdi, year X1** Class #6 / #1 slides **Official solution**

| | |
|---|---|
| Cash received from clients | +180 |
| Cash paid to suppliers | -80 |
| Cash paid to employees | -101 |
| **Operating cash flow** | **-1** |
| Purchase of equipment | -125 |
| **Investing cash flow** | **-125** |
| Proceeds from share issue / bank loan | +150 / +60 |
| Repayment of bank loan / interest paid | -12 / -4 |
| **Financing cash flow** | **+194** |
| Opening 0 → **closing cash** | **68** |

## Reading the statement

- Operating flows: normally positive. Investing: generally negative (positive after a big disposal). Financing: positive or negative depending on the first two.
- Related transactions land in different sections: paying for inventory reduces operating flows; paying for a machine reduces investing flows.
- "Cash is king", but the statement does not tell the whole story.

# Class #7 Introduction to Management Accounting; costing terms and purpose (20 Nov.)

Readings: Horngren, ch. 2, 10 and 3. Cases: Kuniyakino, Montvert, Walk-About. Moodle self-assessment quiz: "The role of management accounting".

**In one look.**

- Costs variable or fixed (by volume); direct or indirect (by cost object).
- = Unit contribution margin = price − variable cost; margin rate = margin / price
- = Break-even point = fixed costs / unit contribution margin
- = Target volume = (fixed costs + target income) / unit contribution margin
- Decide with relevant costs (future, differential); ignore sunk costs.
- Scarce resource: margin per unit of the resource.

## Key concepts · What management accounting is

- It measures and communicates financial and non-financial information to help managers decide (launch a product, accept a client, make or buy…) and shape strategy (cost leadership or differentiation).
- **Financial accounting**: record, report outside, mandatory and standardized, annual, historical. **Management accounting**: specific to each organization, not mandatory, forward-looking, at any frequency (month, week, ad hoc). **Management control** uses this information to steer strategy and operations (budgets, variances, performance measurement).

## Cost terms

- **Cost**: a resource sacrificed to reach a goal. **Cost object**: anything whose cost we want to measure (brand, range, client, project, activity…).
- Not all costs are expenses and vice versa: an investment is neither (cash down, asset up); depreciation is both a cost and an expense with no cash outflow.

| Distinction | Definition | Keep in mind |
|---|---|---|
| **Variable / fixed** | Variable: varies in total with volume. Fixed: does not vary in total over the period despite volume. | Depends on the contract and the horizon: the longer the horizon, the more costs are variable. |
| **Direct / indirect** | Direct: traceable to the cost object in an economically feasible way. Indirect: linked to the object but not traceable; must be allocated. | Depends on the cost object, the materiality of the cost, data collection technology, the organization. |

**Use unit costs cautiously.** Unit cost = total cost / number of units. Kruger: fixed rent 94,500 + handlebar 52 per bike. At 1,000 bikes: (94,500 + 52,000)/1,000 = 146.50. At 600 bikes: (94,500 + 31,200)/600 = 209.50. Using 146.50 would understate the cost.

## Mechanism · The cost-volume-profit (CVP) model

```
Profit = P x Q - v x Q - FC
Contribution margin per unit = P - v
Contribution margin ratio = (P - v) / P
Breakeven volume = FC / (P - v)
Breakeven revenue = FC / margin ratio
Volume for a target profit = (FC + target profit) / (P - v)
Margin of safety = sales - breakeven sales
```

- Assumptions: volume is the only driver; costs split into fixed and variable, linear within the relevant range; price, unit costs and fixed costs known and constant; one product or a constant mix; no time value of money.
- Uses: business plans, feasibility, profitability and risk diagnosis. Limits: single driver, linearity, constant mix, the fixed/variable split depends on the horizon.

### Worked example · Dresses by Mary (official)

**CVP and an advertising decision** Class #7 slides **Official solution**

Purchase 32 + other variable costs 10 per dress, price 70, fixed costs 84,000. Contribution per unit 28 (ratio 40%). At 2,500 dresses: 175,000 - 105,000 - 84,000 = **-14,000**. Breakeven: 84,000/28 = **3,000 dresses**. An advertising campaign of 10,000 lifts sales from 3,200 to 4,000: profit goes from 5,600 to 18,000, i.e. +12,400 → run the campaign.

### Relevant costs for decisions

- **Relevant costs and revenues**: future costs and revenues that **differ** between options. Historical costs only help forecast; **sunk costs** are irrelevant.
- Do not assume all variable costs are relevant, nor that all fixed costs are irrelevant. Full unit costs can mislead (irrelevant costs included; same unit cost at different volumes).
- **Opportunity cost**: the contribution given up by not using a limited resource in its best alternative use; never recorded in accounting.
- **Scarce resource**: at full capacity, rank products by **contribution per unit of the scarce resource**, not per unit of product.
- Pricing: cost is only one input (competition, clients' ability to pay, perceived value); separate special sales from the regular price.

**Noel** (towels): capacity 44,000, production 30,000; variable 8.50 (6.50 + 0.50 + 1.50); fixed 150,000. Full cost (8.5×30,000 + 150,000)/30,000 = 13.50. A special order of 5,000 at 11.50: only variable costs change → contribution 3 × 5,000 = **+15,000**, accept (spare capacity exists).
**Tiilikainen**: motorbike engine, contribution 240 for 2h; boat engine, 375 for 5h; 600h available. Contribution per hour 120 vs 75 → **favor motorbike engines**.

# Class #8 Cost allocation: job order costing and process-costing (27 Nov.)

Readings: Horngren, ch. 4, 5 and 14. Cases: My New Car, Immoservices, Alp'Ski (part 1), WattWatt.

**In one look.**

- = Cost per unit of work = cost centre cost / units of the allocation base
- Job costing (by order) vs process costing (mass production).
- Support centres: direct or sequential method.
- Full cost is no basis for deciding to drop a product.

## Why allocate indirect costs

- Direct costs are traced; indirect costs are shared resources that must be **allocated** to build a full cost (pricing, portfolio strategy, cost reduction, full-cost inventory valuation).

**Careful:** never use a full cost to drop a product or outsource - reason with **differential** costs and revenues instead.

## Key concepts

```
Full cost = traced direct costs + allocated indirect costs
Cost per allocation unit = cost pool total / number of units of the allocation base
Indirect cost charged = units consumed x cost per allocation unit
```

- **Cost pool**: a grouping of indirect costs. **Allocation base** (cost driver, unite d'oeuvre): a measurable variable (hours, kg, euros, m2…) causally linked to the pool's costs. A **homogeneous** pool gathers costs with the same cause.
- Choosing the driver: economic plausibility, quality of fit, regression slope.
- **Support pools** (IT, HR, R&D, maintenance) vs **operating pools** (purchasing, production, distribution).

## Job costing and process costing

| | **Services** | **Manufacturing** |
|---|---|---|
| **Job costing** (cost per order, batch or mission) | Audit, consulting, ad campaign | Aircraft assembly, house building |
| **Process costing** (average cost of identical units) | Deposit processing, standard mail | Refining, food processing |

Job-costing steps: identify the job; its direct costs; the relevant indirect pools; each pool's allocation base; the cost per allocation unit; then charge the job.

## Mechanism · Allocating support pools

1. **Direct method**: only services to operating pools count (simplest, most used).

2. **Step-down method**: allocate first the support pool that serves the other support pools the most, then the next, with no return.
3. **Reciprocal method**: all cross-services (equations); off syllabus. The French "centres d'analyse" method (PCG 1947) uses it.

## Worked example · Direct vs step-down (official)

**Maintenance + IT → Machining, Assembly** Class #8 slides **Official solution**

| | Maintenance | IT | Machining | Assembly |
|---|---|---|---|---|
| Initial costs | 600,000 | 116,000 | 400,000 | 200,000 |
| Maintenance hours | - | 1,600 (20%) | 2,400 (30%) | 4,000 (50%) |
| IT hours | 200 (10%) | - | 1,600 (80%) | 200 (10%) |
| **Direct**: maintenance on 2,400/6,400 and 4,000/6,400; IT on 1,600/1,800 and 200/1,800 | | | 225,000 + 103,111 → **728,111** | 375,000 + 12,889 → **587,889** |
| **Step-down**: maintenance first (20% to IT = 120,000), then IT (236,000) on 1,600/1,800 and 200/1,800 | | | 400,000 + 180,000 + 209,778 → **789,778** | 200,000 + 300,000 + 26,222 → **526,222** |

The method chosen changes the relative cost of products, and therefore performance evaluation. The cost of a product is a **construct**: always state the purpose, the context and the conventions.

# Class #9 Activity-Based Costing (4 Dec.)

Readings: Horngren, ch. 6 and 7. Cases: Alp'Ski (part 2), McCarthy, Castelo Branco.

**In one look.**

- ABC: indirect costs → activities → products via cost drivers.
- = Cost per driver = activity cost / total drivers
- Reveals the true cost of complex products (Alp'Ski, McCarthy, Stella Cuero).
- Limit: heavy database; shifts costs between products.

## Key concepts · Why ABC

- Born in the US in the 1980s, when a single cost pool was no longer enough. Pools are multiplied to stay **homogeneous** and split by **activities** rather than by organizational departments.
- No more support vs operating pools: every activity is charged to cost objects according to its consumption of drivers.

## Mechanism · Implementing ABC

1. Define the cost objects; (2) their direct costs; (3) the activities, their costs and their driver; compute the cost per driver; (4) charge according to driver consumption.

```
Cost per driver = activity cost / total number of drivers
Cost charged to a product = Sum(drivers consumed x cost per driver)
```

## Strengths and limits

- Strengths: faithful modeling (especially of complexity: number of products, components), traceability, well suited to services and processes.
- Limits: a heavy database (drivers x products) to build and maintain, risk of bureaucracy.
- From ABC to **ABM** (activity-based management): cut costs, improve processes, plan activities, set prices and mix, think about customer value. About 40% of US companies use ABC, 20-35% in Europe and France.

Cutting the number of drivers of a product changes the split between products, not necessarily the company's total cost.

# Class #10 Introduction to budget and variance analysis (11 Dec.)

Readings: Horngren, ch. 7 and 8. Cases: Smell Good, Puyricard, Castelo Branco. A mock exam is announced after this class.

**In one look.**

- Budget: a quantified expression of a forward-looking vision over a limited period.
- = Flexible budget = budgeted unit prices and costs × actual volume
- = Sales-volume variance = flexible budget − static budget
- = Flexible-budget variance = actual − flexible budget

## Key concepts · What a budget is

- A quantified expression, over a limited period (usually one year), of a forward-looking view of the company, with financial and non-financial items. A reference point for action. Master budget and departmental budgets; no standard template; most are built on a projected income statement.
- Approach: strategy → objectives → action plans → resources, from past performance and forecasts.
- It sits in a wider process: strategic plan (3-5 years), capital budgets, operational plans (1-3 years), annual budget, departmental budgets (sales, production, materials, labor, overheads, cash, financing).

## Common problems and control

- Imposed without managers, a budget looks unrealistic and is little used; built with them, it takes time and triggers negotiations.
- Control: compare budget and actual, adjust forecasts; this is the pivot of performance and variance analysis.

## Mechanism · Variance analysis (sample exam "A la Coque")

```
Flexible budget = budget recomputed at the actual volume, with budgeted prices and
unit costs
Total (static budget) variance = actual - static budget
Volume variance = flexible budget - static budget
Flexible budget variance = actual - flexible budget
A variance is favorable if it increases profit
```

The slides note that variance analysis was not covered in class, but it appears in a sample paper (A la Coque): know at least these three formulas.

## Course wrap-up

Across Class #7-#10 you covered: the financial accounting / management accounting / control articulation; cost terms and their uses; decision-making with the CVP model; cost allocation methods (direct, step-down, ABC); and budgeting. Management accounting deals with information about the acquisition and consumption of resources, to make managers' decisions easier.

# § Practice, class by class

Every exercise with its full statement and a step-by-step correction. "Official solution" = published by the professor; "Proposed solution" = worked with the course method while the official one is not open on Moodle.

## Class #1 · Overview of financial accounting (9 Oct.) · Practice

Statements are structured from the original course materials; the full narrative wording is in the Workbook and the slide decks on Moodle (source tag on each case).

**Compofruit example** Class #1 slides **Proposed solution**

Compofruit makes jam. Assets or liabilities? A van to carry the fruit; sugar bought but not yet paid for; jam not yet sold; money in the bank; the decision to buy a machine; the employees just hired.

▼ See the solution

The official solution is not published on Moodle yet: this one applies the course method, step by step.

| Item | Treatment |
|---|---|
| Van | Asset (non-current) |
| Unpaid sugar | Asset (raw materials inventory) **and** liability (trade payable) |
| Unsold jam | Asset (finished goods inventory) |
| Money in the bank | Asset (cash) |
| Decision to buy a machine | Nothing: no past event, no controlled resource |
| Employees just hired | Nothing: no recognized asset (no monetary measurement of a controlled resource) |

**Case Verdi (Workbook)** Workbook · solution in the Class #1 slides **Official solution**

Advertising agency created on 1 January X1. (1) Capital contribution 150; (2) bank loan 60; (3) equipment bought for 125, paid by check; (4) campaign for Sam Suffit invoiced 250, payable in 30 days; (5) Sam Suffit pays 180; (6a) salaries paid 101; (6b) interest charged 4; (6c) subcontractor invoice 85 at 60 days; (7) 80 paid to the supplier; (8) 12 repaid on the loan. Build the business equation and the year-end X1 statements.

▼ **See the solution**

| Transaction | Cash | Equipment | Receivables | = Payables | Loans | + Capital | Income |
|---|---|---|---|---|---|---|---|
| 1 Contribution | +150 | | | | | +150 | |
| 2 Loan | +60 | | | | +60 | | |
| 3 Equipment | -125 | +125 | | | | | |
| 4 Service | | | +250 | | | | +250 (sales) |
| 5 Collection | +180 | | -180 | | | | |
| 6a Salaries | -101 | | | | | | -101 |
| 6b Interest | -4 | | | | | | -4 |
| 6c Subcontractor | | | | +85 | | | -85 |
| 7 Payment | -80 | | | -80 | | | |
| 8 Repayment | -12 | | | | -12 | | |
| **Balance** | **68** | **125** | **70** | **5** | **48** | **150** | **60** |

**Total assets 263 = liabilities 53 + equity 210.**

| Balance sheet 31/12/X1 | | Equity and liabilities | |
|---|---|---|---|
| Equipment | 125 | Share capital | 150 |
| Accounts receivable | 70 | Net income | 60 |
| Cash at bank | 68 | Bank loan (non-current) | 48 |
| | | Accounts payable | 5 |
| **Total** | **263** | **Total** | **263** |

| Income statement X1 | |
|---|---|
| Sales | 250 |
| Subcontracting | -85 |
| Personnel expense | -101 |
| Interest | -4 |
| **Net income** | **60** |

| Statement of cash flows X1 | |
|---|---|
| Cash received from clients | +180 |

| | |
|---|---|
| Cash paid to suppliers | -80 |
| Salaries | -101 |
| **Operating cash flow** | **-1** |
| Purchase of equipment | -125 |
| **Investing cash flow** | **-125** |
| Share issue | +150 |
| Bank loan | +60 |
| Repayment | -12 |
| Interest paid | -4 |
| **Financing cash flow** | **+194** |
| **Change in cash (0 → 68)** | **+68** |

Key point: a profit of 60 but an operating cash flow of -1. Here interest is classified in financing.

**Case Busoni (Workbook)** Moodle · Busoni: solution **Official solution**

Company created on 30/09/X1. (1) Contribution of equipment 100 and 40 in cash; (2) goods bought on credit 60; (3) loan 100; (4) services on credit 50; (5) salaries paid 20; (6) client collection 42; (7) credit sale, 40, of half of the goods; (8) electricity bill 15; (9) suppliers paid 45; (10) repayment 10. Business equation, income statement and balance sheet.

▼ See the solution

| Transaction | Equip. | Cash | Goods | Receivables | = Payables | Loan | + Capital | Income |
|---|---|---|---|---|---|---|---|---|
| 1 | +100 | +40 | | | | | +140 | |
| 2 | | | +60 | | +60 | | | |
| 3 | | +100 | | | | +100 | | |
| 4 | | | | +50 | | | | +50 |
| 5 | | -20 | | | | | | -20 |
| 6 | | +42 | | -42 | | | | |
| 7 sale | | | | +40 | | | | +40 |
| 7 cost of sales | | | -30 | | | | | -30 |
| 8 | | | | | +15 | | | -15 |
| 9 | | -45 | | | -45 | | | |
| 10 | | -10 | | | | -10 | | |
| **Balance** | **100** | **107** | **30** | **48** | **30** | **90** | **140** | **25** |

Total assets 285 = liabilities 120 + equity 165. The credit sale (7) happens in two steps: revenue of 40, then a "cost of goods sold" expense of 60/2 = 30.

| **Income statement** | |
|---|---|
| Services | 50 |
| Sales of goods | 40 |
| Cost of sales | -30 |
| Salaries | -20 |
| Electricity | -15 |
| **Net income** | **25** |

| **Balance sheet** | | | |
|---|---|---|---|
| Equipment | 100 | Share capital | 140 |
| Goods | 30 | Net income | 25 |
| Receivables | 48 | Bank loan (non-current) | 90 |
| Cash | 107 | Accounts payable | 30 |
| **Total** | **285** | **Total** | **285** |

## Class #2 · Operating cycle and current assets (16 Oct.) · Practice

Statements are structured from the original course materials; the full narrative wording is in the Workbook and the slide decks on Moodle (source tag on each case).

### Case Cameras (slides) Class #2 slides Official solution

A shop buys and re-sells one single type of video cameras. The January opening inventory consisted in 50 video cameras valued at 100 euros each. The following transactions occurred during January: 10.01: acquires 75 video cameras at 110 euros each; 15.01: sells 60 video cameras at 150 euros each; 20.01: acquires 25 video cameras at 120 euros each; 30.01: sells 40 video cameras at 150 euros each. Compute the value of the inventory at the end of January based on each of the three methods (WAC, FIFO, LIFO). What is the impact of each valuation method on the income statement and the balance sheet? Assume all transactions are paid in cash.

▼ See the solution

**WAC**: on 15/01 the average is (50×100 + 75×110)/125 = 106 → COGS 60×106 = 6,360, inventory 65×106 = 6,890. On 30/01: (65×106 + 25×120)/90 = 109.9 → COGS 40×109.9 = 4,396; closing inventory 50×109.9 = 5,495.

**FIFO**: on 15/01 out go 50×100 + 10×110 = 6,100; on 30/01 out go 40×110 = 4,400; closing inventory 25×110 + 25×120 = 5,750.

**LIFO**: on 15/01 out go 60×110 = 6,600; on 30/01 out go 25×120 + 15×110 = 4,650; closing inventory 50×100 = 5,000.

| | FIFO | WAC | LIFO |
|---|---|---|---|
| Sales | 15,000 | 15,000 | 15,000 |
| COGS | 10,500 | 10,756 | 11,250 |
| Profit | 4,500 | 4,244 | 3,750 |
| Closing inventory | 5,750 | 5,495 | 5,000 |

**Case Mozart (Workbook)** Workbook **Proposed solution**

A single-product shop opened on 1 April. 01/04 contribution of 5,000; 02/04 buy 200 units at 22, paid cash; 08/04 buy 180 units at 18 on credit (end of May); 17/04 sell 240 units at 25 cash; 22/04 April rent 500; 25/04 sell 20 units at 26 on credit; 27/04 salaries 400; 30/04 pay the month's tax (40%). Balance sheet, income statement and cash flows under FIFO, LIFO and WAC, then a comparison.

### ▼ See the solution

The official solution is not published on Moodle yet: this one applies the course method, step by step.

**Cost of sales** (260 units sold, 120 in stock).

- FIFO: 17/04 → 200×22 + 40×18 = 5,120; 25/04 → 20×18 = 360; total 5,480; closing inventory 120×18 = 2,160.
- LIFO: 17/04 → 180×18 + 60×22 = 4,560; 25/04 → 20×22 = 440; total 5,000; closing inventory 120×22 = 2,640.
- WAC: average after 08/04 = 7,640/380 = 20.105; cost of sales 260×20.105 = 5,227.37; closing inventory 120×20.105 = 2,412.63.

| Income statement | WAC | FIFO | LIFO |
|---|---|---|---|
| Sales (6,000 + 520) | 6,520 | 6,520 | 6,520 |
| Cost of sales | -5,227.37 | -5,480 | -5,000 |
| Gross margin | 1,292.63 | 1,040 | 1,520 |
| Rent | -500 | -500 | -500 |
| Salaries | -400 | -400 | -400 |
| Profit before tax | 392.63 | 140 | 620 |
| Tax 40% | -157.05 | -56 | -248 |
| **Net income** | **235.58** | **84** | **372** |

| Balance sheet 30/04 | WAC | FIFO | LIFO |
|---|---|---|---|
| Goods | 2,412.63 | 2,160 | 2,640 |
| Receivables | 520 | 520 | 520 |
| Cash | 5,542.95 | 5,644 | 5,452 |
| **Total assets** | **8,475.58** | **8,324** | **8,612** |
| Capital | 5,000 | 5,000 | 5,000 |
| Net income | 235.58 | 84 | 372 |
| Accounts payable | 3,240 | 3,240 | 3,240 |
| **Total equity+liabilities** | **8,475.58** | **8,324** | **8,612** |

| Cash flows | WAC | FIFO | LIFO |
|---|---|---|---|
| Cash from clients | +6,000 | +6,000 | +6,000 |
| Paid to suppliers | -4,400 | -4,400 | -4,400 |

| | | | |
|---|---|---|---|
| Rent | -500 | -500 | -500 |
| Salaries | -400 | -400 | -400 |
| Tax paid | -157.05 | -56 | -248 |
| Shareholders' contribution | +5,000 | +5,000 | +5,000 |
| **End-of-April cash** | **5,542.95** | **5,644** | **5,452** |

Here purchase prices **fall** (22 then 18): LIFO gives the highest profit and FIFO the lowest, the opposite of the Cameras case. The only cash difference comes from tax.

**Schultz exercise (revenue recognition)** Moodle · Additional exercise Schultz **Official solution**

(a) A school invoices on 1 December X1 the fees for the first term of X2, all collected on 15/12, and recognizes revenue on collection. (b) A shipyard builds a cargo ship (started July X2, 10% complete at end X2, 80% at end X3, delivered April X4; estimated cost 80M) and recognizes revenue on delivery. (c) A bridge club spreads membership fees linearly over the membership period. (d) A shipper entrusts goods to a consignee who sells them on its behalf, and recognizes revenue on delivery to the consignee.

▼ See the solution

- (a) Wrong: the revenue belongs to X2 (deferred revenue), even if the cash is collected in X1.
- (b) Revenue should rather be recognized **over time** (percentage of completion), since revenue, costs and progress are reliably measurable.
- (c) Correct.
- (d) Wrong: revenue is recognized when the consignee sells the goods; unsold goods return to the shipper.

**Bruckner example (doubtful receivable)** Class #2 slides **Proposed solution**

End of X1, Bruckner holds 100 on client Anton, in difficulty: only 60 is expected to be collected. End of X2, the expected loss rises to 70. In X3, Anton is liquidated and pays only 10. Impact in X1, X2 and X3?

▼ See the solution

The official solution is not published on Moodle yet: this one applies the course method, step by step.

| Year | Balance sheet (net receivable) | Income statement |
|---|---|---|
| X1 | 100 - allowance 40 = 60 | Impairment expense -40 |
| X2 | 100 - allowance 70 = 30 | Additional charge -30 |
| X3 | Receivable written off; 10 collected | Total loss 90, of which 70 already covered: net charge -20 |

## Class #3 · Investing cycle and non-current assets (23 Oct.) · Practice

Statements are structured from the original course materials; the full narrative wording is in the Workbook and the slide decks on Moodle (source tag on each case).

**Slide examples: straight-line and declining balance** Class #3 slides **Official solution**

(1) Cost 11,000, residual value 1,000, 5 years, straight-line. (2) NCA machine of 15,000 over 10 years, double declining balance: charges for the first two years.

▼ See the solution

| Year-end | Charge | Cumulated | NBV |
|---|---|---|---|
| X1 | 2,000 | 2,000 | 9,000 |
| X2 | 2,000 | 4,000 | 7,000 |
| X3 | 2,000 | 6,000 | 5,000 |
| X4 | 2,000 | 8,000 | 3,000 |
| X5 | 2,000 | 10,000 | 1,000 |

(1) Depreciable amount 10,000 × 20% = 2,000 per year. (2) Straight-line rate 10% → DDBM 20%. Year 1: 15,000 × 20% = 3,000 (NBV 12,000). Year 2: 12,000 × 20% = 2,400; cumulated 5,400; NBV 9,600.

**Case Amorti (Workbook)** Workbook · full official Amorti solution also on Moodle **Proposed solution**

Five-year project: manufacturing licence 75,000, production line 300,000, launch advertising campaign 100,000. Expected sales 1,000,000 per year, other expenses 800,000. Income statements of the first two years (a) straight-line and (b) 40% declining balance for the line. Are the licence and the advertising assets or expenses?

▼ **See the solution**

The official solution is not published on Moodle yet: this one applies the course method, step by step.

**Licence**: an acquired finite-life intangible asset → amortized over 5 years, 15,000 per year. **Advertising**: never an asset under the course rules → expense of 100,000 in year 1. **Line**: straight-line 300,000/5 = 60,000 per year; declining 40%: 120,000 then 72,000 (40% × 180,000).

| | SL Y1 | SL Y2 | DDB Y1 | DDB Y2 |
|---|---|---|---|---|
| Sales | 1,000,000 | 1,000,000 | 1,000,000 | 1,000,000 |
| Other expenses | -800,000 | -800,000 | -800,000 | -800,000 |
| Licence amortization | -15,000 | -15,000 | -15,000 | -15,000 |
| Advertising | -100,000 | 0 | -100,000 | 0 |
| Line depreciation | -60,000 | -60,000 | -120,000 | -72,000 |
| **Profit** | **25,000** | **125,000** | **-35,000** | **113,000** |

Over 5 years the total is identical (525,000): the method only changes the timing. DDB years 3 to 5: 43,200, 25,920, then the remaining 38,880.

## Class #4 · Financing cycle, equity and liabilities (30 Oct.) · Practice

Statements are structured from the original course materials; the full narrative wording is in the Workbook and the slide decks on Moodle (source tag on each case).

**Case ACME (Workbook)** Workbook · official ACME solution opens on Moodle 31 Oct, 9:00 **Proposed solution**

ACME was created with share capital of 5,000 k€, par value 25 per share. It needs 5,000 k€ through a capital increase: the share price is 110, the new shares are issued at 100. Current equity: capital 5,000, reserves (retained earnings) 1,500, net income 800 (k€). (1) Number of shares at creation; (2) number of new shares; (3) equity after the operation.

### ▼ See the solution

The official solution is not published on Moodle yet: this one applies the course method, step by step.

1. 5,000,000 / 25 = **200,000 shares**.
2. 5,000,000 / 100 = **50,000 new shares**.
3. Capital: +50,000 × 25 = +1,250 k€; share premium: +50,000 × (100 - 25) = +3,750 k€; cash +5,000 k€.

| Equity (k€) | Before | After |
|---|---|---|
| Share capital | 5,000 | 6,250 |
| Share premium | 0 | 3,750 |
| Reserves | 1,500 | 1,500 |
| Net income | 800 | 800 |
| **Total** | **7,300** | **12,300** |

The issue price is below the market price (100 vs 110) to encourage subscription. Book value per share after: 12,300 k€ / 250,000 = 49.2.

**Gerry Co exercise (profit appropriation)** Moodle · Additional exercise Gerry Co. Official solution

Equity at end of X1: capital 100, premium 18, reserves 50 (total 168). Net income X2: 70; no reserve, dividends declared 30. Present X2 equity before and after appropriation.

▼ See the solution

| Before appropriation | X1 | X2 |
|---|---|---|
| Capital | 100 | 100 |
| Share premium | 18 | 18 |
| Retained earnings (opening) | 0 | 50 |
| Income to appropriate | 50 | 70 |
| **Equity** | **168** | **238** |

| After appropriation | X1 | X2 |
|---|---|---|
| Capital | 100 | 100 |
| Share premium | 18 | 18 |
| Reserves (closing) | 50 | 90 = 50 + 70 - 30 |
| **Equity** | **168** | **208** |
| Dividends payable (liability) | 0 | 30 |
| **Total** | **168** | **238** |

## Class #5 · End-of-Year adjustments (6 Nov.) · Practice

Statements are structured from the original course materials; the full narrative wording is in the Workbook and the slide decks on Moodle (source tag on each case).

**Litigation example (slides)** Class #5 slides Proposed solution

A client sues the company in 20X1 for 10M. End 20X1: losing is unlikely. End 20X2: a probable loss of 8M. End 20X3: probable amount 9M. In 20X4: settled for 9.5M. Impacts?

▼ See the solution

The official solution is not published on Moodle yet: this one applies the course method, step by step.

| Year | Balance sheet | Income statement |
|---|---|---|
| 20X1 | Nothing (contingent liability in the notes) | - |
| 20X2 | Provision 8M | Expense -8M |
| 20X3 | Provision 9M | Additional expense -1M |
| 20X4 | Provision used (-9M), cash -9.5M | Additional expense -0.5M |

**Case Flossie Flowers (Workbook)** Workbook **Proposed solution**

Entries at 31/12/X2. (A) Flowers delivered on 27/12/X2, the 2,400 invoice sent on 05/01/X3. (B) Internet access bill of 01/12/X2 covering December and January, 200, paid on 15/12. (C) Estimated December expenses: phone 480 (administration), electricity and heating 620 (distribution), bills arriving in January. (D) Loan of 100,000 from 01/01/X1 at 10%, repaid in constant annual instalments of 26,380 each January; nothing recorded for X2, of which principal 18,018 and interest 8,362.

**▼ See the solution**

The official solution is not published on Moodle yet: this one applies the course method, step by step.

| | Assets | Liabilities | Equity (income) |
|---|---|---|---|
| A | Accrued revenue (unbilled) +2,400 | | Sales +2,400 |
| B | Prepaid expense +100 (January) | | Internet expense -100 (the X2 expense is only 100) |
| C | | Accrued expenses +1,100 | Phone -480; electricity and heating -620 |
| D | | Accrued interest +8,362; 18,018 of the loan reclassified as current | Interest expense -8,362 |

Check D: X1 interest = 10% × 100,000 = 10,000, principal repaid 16,380, balance 83,620; X2 interest = 8,362, principal 18,018 (total 26,380).

**Case Imp (Workbook)** Workbook **Proposed solution**

Machine bought on 02/01/20X3 for 3,000,000, depreciated straight-line over 5 years (600,000 per year). Impairment test at end 20X6: useful life unchanged, value in use 800,000, net selling price 950,000. Accounting effects in 20X6?

**▼ See the solution**

The official solution is not published on Moodle yet: this one applies the course method, step by step.

After the 20X6 charge (4th year), net book value = 3,000,000 - 4 × 600,000 = **600,000**. Recoverable amount = max(950,000; 800,000) = **950,000**. It exceeds the carrying amount: **no impairment loss**. Only the normal 600,000 charge is recorded, and the asset is not revalued upwards.

If the test were done before the 20X6 charge (NBV 1,200,000), the loss would be 1,200,000 - 950,000 = 250,000. Always check the date of the test.

**Case Provisor (Workbook)** Workbook **Proposed solution**

Provision or not? (a) One-year warranty, 3,000,000 of sales, 1-3% of products defective and replaced. (b) Phosphate extraction on own land, estimated restoration costs 5,000,000. (c) Lawsuit, loss judged probable, 350,000. (d) Airline legally required to overhaul its planes every three years, future cost estimated at 2,500,000.

▼ **See the solution**

The official solution is not published on Moodle yet: this one applies the course method, step by step.

- (a) **Yes**: present obligation (past sales), probable outflow, estimable: between 30,000 and 90,000, e.g. 60,000 (2%) from experience.
- (b) **Yes only if a legal or constructive obligation** to restore exists (law, public commitment). Otherwise no provision (like the slide's oil company while no law exists).
- (c) **Yes**: provision of 350,000.
- (d) **No**: the overhaul relates to future operation of the planes, not a past event; the company could sell the planes. No present obligation, so no provision (and future operating losses or expenses are never provided for).

**Offenbach exercise (type of adjustment)** Moodle · Additional exercise Offenbach (statement only) **Proposed solution**

For each case, choose: prepaid expense, deferred revenue, accrued expense, accrued revenue. (1) Four-month loan granted, one month of interest accrued, collected at maturity (March X2). (2) Office rented on 01/04/X1, one year of rent collected. (3) Loan received on 01/10/X1, repayable September X2, 10% interest paid at maturity. (4) Six-month insurance paid on 01/09/X1. (5) Year-end bonuses decided in X1, unpaid. (6) Advertising run in late December X1, invoice not received.

▼ **See the solution**

The official solution is not published on Moodle yet: this one applies the course method, step by step.

| Case | Answer |
|---|---|
| 1 | Accrued revenue (accrued interest) |
| 2 | Deferred revenue (3 months of X2) |
| 3 | Accrued expense (3 months of interest) |
| 4 | Prepaid expense (January and February X2) |
| 5 | Accrued expense |
| 6 | Accrued expense |

## Class #6 · Statement of cash flows (13 Nov.) · Practice

Statements are structured from the original course materials; the full narrative wording is in the Workbook and the slide decks on Moodle (source tag on each case).

**Case Beethoven (Workbook)** Workbook **Proposed solution**

Balance sheet at 31/12/X1 (k€): equipment 900; inventories 200, receivables 220, bank 140; capital 400, reserves 250, X1 income 360; loans 120, trade payables 90, tax payable 240. In X2: (1) sales 1,600 of which 1,400 collected; (2) purchases 500 of which 480 paid; (3) repayment 50, new loan 70, interest paid 45; (4) salaries 450; (5) rent 300; (6) other expenses 75; (7) equipment 200 paid; (8) depreciation 50; (9) capital increase 240 at par; (10) closing inventory 220. Tax 40%, paid in X3. Statement of cash flows, income statement and end-of-X2 balance sheet.

## ▼ See the solution

The official solution is not published on Moodle yet: this one applies the course method, step by step.

Excel model assumptions: X1 income goes to reserves; X1 receivables, trade payables and tax are settled in X2. Cost of sales = 200 + 500 - 220 = 480.

| **Income statement X2** | **k€** |
|---|---|
| Sales | 1,600 |
| Cost of sales | -480 |
| Salaries | -450 |
| Rent | -300 |
| Other expenses | -75 |
| Depreciation | -50 |
| Interest | -45 |
| Profit before tax | 200 |
| Tax 40% | -80 |
| **Net income** | **120** |

| **Statement of cash flows X2** | **k€** |
|---|---|
| Collected from clients (220 + 1,400) | +1,620 |
| Suppliers (90 + 480) | -570 |
| X1 tax | -240 |
| Salaries | -450 |
| Rent | -300 |
| Other expenses | -75 |
| Interest | -45 |
| **Operating cash flow** | **-60** |
| Equipment purchase | -200 |
| **Investing cash flow** | **-200** |
| Capital increase | +240 |
| New loan | +70 |

| | |
|---|---|
| Repayment | -50 |
| **Financing cash flow** | **+260** |
| **Change (140 → 140)** | **0** |

| **Balance sheet 31/12/X2** | | | |
|---|---|---|---|
| Equipment (900 + 200 - 50) | 1,050 | Capital (400 + 240) | 640 |
| Inventories | 220 | Reserves (250 + 360) | 610 |
| Receivables (1,600 - 1,400) | 200 | X2 income | 120 |
| Bank | 140 | Loans (120 - 50 + 70) | 140 |
| | | Trade payables (500 - 480) | 20 |
| | | Tax payable | 80 |
| **Total** | **1,610** | **Total** | **1,610** |

A profit of 120, but a negative operating cash flow (-60): the X1 tax payment and partial collection of sales weigh on cash. Here interest is classified as operating, as in the Excel model.

**Indirect method: a numeric example** Applying the Class #6 slides **Proposed solution**

Net income 100; depreciation 50; trade receivables +30; trade payables +20. Operating cash flow?

**▼ See the solution**

The official solution is not published on Moodle yet: this one applies the course method, step by step.

100 + 50 - 30 + 20 = **140**. Depreciation is added back (a non-cash expense); the increase in receivables is subtracted (uncollected sales); the increase in payables is added (unpaid purchases).

## Class #7 · Introduction to Management Accounting; costing terms and purpose (20 Nov.) · Practice

Statements are structured from the original handouts; the full original wording of each case is preserved in its Moodle entry on the Exercices tab.

**Kuniyakino (cost terms)** Official Kuniyakino solution opens on Moodle 21 Nov, 12:00 **Proposed solution**

A Singapore bookshop, Management department. Classify each cost as direct or indirect (relative to the department) and variable or fixed (relative to the number of books sold): 1. shop electricity (single bill); 2. books bought for resale; 3. subscription to the magazine *Business Books for leaders*; 4. accounting software rental; 5. store manager's salary; 6. commission on the department head's sales; 7. free bookmarks (one per customer); 8. metro advertising; 9. fire insurance; 10. merchandising for the launch of *Innovative Strategies*.

### ▼ See the solution

The official solution is not published on Moodle yet: this one applies the course method, step by step.

| Cost | D / I | V / F |
|---|---|---|
| 1 Electricity | I | F |
| 2 Books bought | D | V |
| 3 Magazine subscription | D | F |
| 4 Accounting software | I | F |
| 5 Store manager | I | F |
| 6 Department head's commission | D | V |
| 7 Bookmarks | I | V |
| 8 Metro advertising | I | F |
| 9 Fire insurance | I | F |
| 10 Book merchandising | D | F |

The bookmark is given to all shop customers (indirect for the department) but its cost follows the number of customers (variable).

**Montvert (breakeven)** Official Montvert solution opens on Moodle 21 Nov, 19:00 **Proposed solution**

Pens sold at 2; 2024 sales: 20M euros; fixed costs 3,600,000; variable cost 1.20. (1) Current profit and breakeven in volume and revenue. (2) Independent questions: (a) variable cost +0.16; (b) advertising +360,000 of fixed costs and +10% volume; (c) 20% discount, volume +40%, variable cost -10%; (d) move upmarket: price +10%, fixed costs +80,000.

### ▼ See the solution

The official solution is not published on Moodle yet: this one applies the course method, step by step.

Volume 20M / 2 = 10M units. Unit margin 2 - 1.20 = 0.80.

| Case | Unit margin | Volume | Fixed | Profit | Breakeven (units) | Breakeven (revenue) |
|---|---|---|---|---|---|---|
| Current | 0.80 | 10M | 3.6M | **4.4M** | 4.5M | 9M |
| (a) | 0.64 | 10M | 3.6M | 2.8M | 5.625M | 11.25M |
| (b) | 0.80 | 11M | 3.96M | 4.84M | 4.95M | 9.9M |
| (c) price 1.60, v 1.08 | 0.52 | 14M | 3.6M | 3.68M | ≈ 6.92M | ≈ 11.08M |
| (d) price 2.20 | 1.00 | 10M (unchanged) | 3.68M | 6.32M | 3.68M | 8.10M |

Moving upmarket (d) maximizes profit if volume holds; the promotion (c) raises the breakeven sharply.

**Walk-About (CVP)** Official Walk-About solution opens on Moodle 21 Nov, 19:00 **Proposed solution**

Shoe shop: price 30 per pair, 5% commission, shoe cost 19.50; fixed: rent 60,000, salaries 200,000, advertising 80,000, other 20,000. (1) Breakeven in units and revenue; profit at 35,000 pairs. (2) If the manager gets 0.30 per pair. (3) Commission removed against +81,000 of fixed salaries: (a) breakeven; (b) indifference volume between the two plans; (c) profits at 50,000 and 60,000 pairs; (d) volume for a target profit of 168,000.

### ▼ See the solution

The official solution is not published on Moodle yet: this one applies the course method, step by step.

Variable cost 19.50 + 1.50 = 21; margin 9; fixed 360,000.

1. Breakeven 360,000/9 = **40,000 pairs**, i.e. 1,200,000. At 35,000: 35,000 × 9 - 360,000 = **-45,000**.
2. Margin 8.70 → breakeven 41,379.3, rounded to **41,380 pairs** (1,241,400).
3. (a) Margin 10.50, fixed 441,000 → breakeven **42,000 pairs** (1,260,000). (b) 10.5Q - 441,000 = 9Q - 360,000 → **Q = 54,000**. (c) 50,000 pairs: fixed 84,000, commission 90,000; 60,000 pairs: fixed 189,000, commission 180,000. Beyond 54,000, the fixed plan pays more. (d) Fixed: (441,000 + 168,000)/10.5 = **58,000**; commission: (360,000 + 168,000)/9 = **58,667** pairs.

**My new car (relevant costs)** Official My new car solution opens on Moodle 28 Nov, 19:00 **Proposed solution**

New car for 19,750, 25,000 km a year of personal use. The employer offers 0.35/km for 15,000 business km a year. The car is changed around 75,000 to 80,000 km (trade-in at market). Resale values: 15,200 (1 yr), 14,350 (2 yrs), 13,450 (3 yrs), 12,100 (4 yrs), 10,200 (5 yrs). Insurance 600/yr (private) or 900/yr (private + business); parking 600/yr; 9 l/100 km at 1.40/l; maintenance 0.10/km. Accept?

### ▼ See the solution

The official solution is not published on Moodle yet: this one applies the course method, step by step.

| Annual cost | Without business (25,000 km) | With business (40,000 km) |
|---|---|---|
| Value loss | Resale after 3 yrs (75,000 km): 6,300/3 = 2,100 | Resale after 2 yrs (80,000 km): 5,400/2 = 2,700 |
| Insurance | 600 | 900 |
| Parking | 600 | 600 |
| Fuel (0.126/km) | 3,150 | 5,040 |
| Maintenance | 2,500 | 4,000 |
| **Total** | **8,950** | **13,240** |

Differential cost: 4,290 for 15,000 km, i.e. **0.286/km** < 0.35: **accept** (a gain of about 960 a year). Parking is not relevant (identical in both cases); the purchase price is already paid, only the resale value difference matters.

## Class #8 · Cost allocation: job order costing and process-costing (27 Nov.) · Practice

Statements are structured from the original handouts; the full original wording of each case is preserved in its Moodle entry on the Exercices tab.

**Immoservices (job costing)** Official Immoservices solution not open on Moodle yet **Proposed solution**

Real-estate consulting firm, price = cost + 30%. Direct labor: 70/h; a single indirect pool (21,000) allocated per professional labor-hour. Pemberley: 104 h; Netherfield: 96 h. (1) Why is it important for Immoservices to understand the costs associated with individual cases? Present an overview diagram of the existing job-costing system (one direct-cost category: professional labor; one indirect-cost pool: general support, allocated per labor-hour). (2) Cost and price of each engagement. (3) The pool is split: administration 7,000 (per hour) and documentation 14,000 (per request: Pemberley 2, Netherfield 8).

▼ See the solution

**1. Why it matters, and the current system.** Knowing the cost of each case lets Immoservices defend its prices to clients (Mr Darcy's complaint), bid against competitors on the next job, and see which cases are really profitable. Current system: one direct-cost category (professional labor at 70/h) + one indirect-cost pool (general support, 21,000) allocated to cases per professional labor-hour.

The official solution is not published on Moodle yet: this one applies the course method, step by step.

Q1: knowing the cost per engagement supports competitive pricing, client profitability analysis and spotting inefficiencies. Scheme: direct costs (hours × 70) + "general support" pool allocated per hour → engagement cost → price (+30%).

| | Pemberley | Netherfield |
|---|---|---|
| Labor (h × 70) | 7,280 | 6,720 |
| Indirect, 1 pool (105/h = 21,000/200) | 10,920 | 10,080 |
| **Cost** | **18,200** | **16,800** |
| **Price (+30%)** | **23,660** | **21,840** |
| Administration (35/h) | 3,640 | 3,360 |
| Documentation (1,400/request) | 2,800 | 11,200 |
| **New cost** | **13,720** | **21,280** |
| **New price** | **17,836** | **27,664** |

Pemberley was overcharged: with two pools its price falls by nearly 5,800, which answers Mr Darcy's complaint.

**Alp'Ski, part 1 (cost pools)** Official Alp Ski (part 1) solution not open on Moodle yet **Proposed solution**

GRIMPER skis (800 pairs at 500) and EXTREME (210 pairs at 2,000). Direct costs: GRIMPER 250,000 (materials 80,000; cutting labor 500 h and assembly 2,000 h at 40; other fixed 70,000); EXTREME 300,000 (materials 220,000; cutting 500 h, assembly 500 h; other fixed 40,000). Indirect: cutting 35,000, assembly 45,000, IT 60,000, purchasing 25,000, maintenance 50,000. Maintenance allocated 20% cutting, 20% assembly, 30% IT, 30% purchasing; then IT and purchasing 40% cutting, 60% assembly. Cutting charged per k€ of materials; assembly per assembly labor hour. Total and per-pair cost.

### ▼ See the solution

The official solution is not published on Moodle yet: this one applies the course method, step by step.

| Allocation | Cutting | Assembly | IT | Purchasing | Maintenance |
|---|---|---|---|---|---|
| Initial costs | 35,000 | 45,000 | 60,000 | 25,000 | 50,000 |
| Maintenance | +10,000 | +10,000 | +15,000 | +15,000 | -50,000 |
| IT (75,000) | +30,000 | +45,000 | -75,000 | | |
| Purchasing (40,000) | +16,000 | +24,000 | | -40,000 | |
| **Total** | **91,000** | **124,000** | 0 | 0 | 0 |
| Allocation unit | 300 k€ of materials → 303.33/k€ | 2,500 h → 49.60/h | | | |

| | GRIMPER | EXTREME |
|---|---|---|
| Direct costs | 250,000 | 300,000 |
| Cutting | 80 × 303.33 = 24,267 | 220 × 303.33 = 66,733 |
| Assembly | 2,000 × 49.60 = 99,200 | 500 × 49.60 = 24,800 |
| **Total cost** | **373,467** | **391,533** |
| Cost per pair | 466.83 | 1,864.44 |
| Profit | +26,533 (33.17/pair) | +28,467 (135.56/pair) |

**WattWatt (choosing the allocation base)** Official WattWatt solution not open on Moodle yet **Proposed solution**

Engineering consultancy. Engineer labor: 1,500,000 SGD; overheads 2,400,000. Contracts Alpha (labor 300,000, 6,000 h, 75 days) and Beta (75,000, 1,000 h, 30 days). Total hours 25,600; mission days 3,650. (1) Allocate on labor cost; (2) on hours; (3) the consultant's method: expert fees (300,000), photocopies (100,000) and travel (318,000) become direct (Alpha: 20,000, 10,000, 20,000; Beta: 80,000, 30,000, 10,000); remaining telecoms 401,500 (days), HR 512,000 (hours), other 768,500 (labor cost).

**▼ See the solution**

The official solution is not published on Moodle yet: this one applies the course method, step by step.

| | Alpha | Beta |
|---|---|---|
| (1) Rate 2,400,000/1,500,000 = 1.6 per SGD | 300,000 + 480,000 = **780,000** | 75,000 + 120,000 = **195,000** |
| (2) Rate 2,400,000/25,600 h = 93.75/h | 300,000 + 562,500 = **862,500** | 75,000 + 93,750 = **168,750** |
| (3) Direct costs | 300,000 + 50,000 = 350,000 | 75,000 + 120,000 = 195,000 |
| Telecoms (110 per day) | 8,250 | 3,300 |
| HR (20 per hour) | 120,000 | 20,000 |
| Other (0.5123 per SGD of labor) | 153,700 | 38,425 |
| **(3) Total cost** | **631,950** | **256,725** |

The third method is the most faithful: more costs traced directly and drivers linked to causes. Beta, a small contract but a heavy consumer of experts, was strongly under-costed by the first two methods.

## Class #9 · Activity-Based Costing (4 Dec.) · Practice

Statements are structured from the original handouts; the full original wording of each case is preserved in its Moodle entry on the Exercices tab.

**Alp'Ski, part 2 (ABC)** Official Alp Ski solution not open on Moodle yet **Proposed solution**

Eleven activities (215,000 in total). Purchasing 30,000 (10 orders: G 7, E 3); quality control 25,000 (25 batches: 15, 10); transfers 10,000 (15: 10, 5); wood cutting 10,000 (1,515 kg: 1,200, 315); resin cutting 20,000 (240 kg: 240, 0); fiber cutting 10,000 (547 kg: 400, 147); gluing 10,000 (5 materials: 3, 2); drying 30,000 (1,852 h: 1,600, 252); programming 20,000 (820 lines: 400, 420); setups 40,000 (25: 20, 5); IT support 10,000 (40 requests: 25, 15). Profit of each ski?

### ▼ See the solution

The official solution is not published on Moodle yet: this one applies the course method, step by step.

| Activity | Cost per driver | GRIMPER | EXTREME |
|---|---|---|---|
| Purchasing | 3,000 | 21,000 | 9,000 |
| Quality | 1,000 | 15,000 | 10,000 |
| Transfers | 666.67 | 6,667 | 3,333 |
| Wood | 6.60 | 7,921 | 2,079 |
| Resin | 83.33 | 20,000 | 0 |
| Fiber | 18.28 | 7,313 | 2,687 |
| Gluing | 2,000 | 6,000 | 4,000 |
| Drying | 16.20 | 25,918 | 4,082 |
| Programming | 24.39 | 9,756 | 10,244 |
| Setups | 1,600 | 32,000 | 8,000 |
| IT support | 250 | 6,250 | 3,750 |
| **Total indirect** | | **157,824** | **57,176** |
| Direct costs | | 250,000 | 300,000 |
| **Profit** | | **-7,824** (-9.78/pair) | **+62,824** (+299.16/pair) |

With ABC, GRIMPER (complex product, small batches, many customers) turns loss-making, although the pool method showed it profitable. EXTREME is far more profitable.

**McCarthy (ABC)** Official McCarthy solution not open on Moodle yet **Proposed solution**

Frozen fries: 1,000,000 kg (900,000 retail, 100,000 catering). Potatoes 150,000; production support 983,000 allocated per kg. (1) Cost per kg. (2) 188,000 of packaging becomes direct (180,000 retail, 8,000 catering), the rest per kg. (3) Activities: washing 120,000 (kg); cutting 231,000 (hours: 250 kg/h retail, 400 kg/h catering); packing 444,000 (hours: 25 kg/h retail, 100 kg/h catering). Cost per driver, cost per kg per market, use of the information.

### ▼ See the solution

The official solution is not published on Moodle yet: this one applies the course method, step by step.

1. (150,000 + 983,000)/1,000,000 = **1.133/kg** for everyone.
2. Remaining 795,000/1,000,000 = 0.795/kg. Retail: 0.15 + 0.795 + 180,000/900,000 (0.20) = **1.145**; catering: 0.15 + 0.795 + 0.08 = **1.025**.
3. Washing 0.12/kg. Cutting: 3,600 h retail + 250 h catering = 3,850 h → **60/h**. Packing: 36,000 h + 1,000 h = 37,000 h → **12/h**.

| Cost per kg | Retail | Catering |
|---|---|---|
| Potatoes | 0.15 | 0.15 |
| Direct packaging | 0.20 | 0.08 |
| Washing | 0.12 | 0.12 |
| Cutting | 216,000/900,000 = 0.24 | 15,000/100,000 = 0.15 |
| Packing | 432,000/900,000 = 0.48 | 12,000/100,000 = 0.12 |
| **Total** | **1.19** | **0.62** |

The catering market costs 0.62/kg, not 1.133: the old system overcharged it, hence a bid priced 30% too high. Uses: pricing tenders, market profitability analysis, reducing activity costs (ABM).

## Class #10 · Introduction to budget and variance analysis (11 Dec.) · Practice

Statements are structured from the original handouts; the full original wording of each case is preserved in its Moodle entry on the Exercices tab.

**Castelo Branco (budget)** Official Castelo Branco solution not open on Moodle yet **Proposed solution**

Estimated 2024 profit (k€): equipment sales 6,000, maintenance contracts 1,800; cost of sales 4,600; marketing 600, distribution 150, client maintenance 1,000, administration 900; profit 550. 2025 assumptions: equipment prices +10%, contract prices unchanged; volumes +6% (equipment and contracts); unit cost +3%; marketing +250, administration stable; distribution proportional to units; two maintenance technicians for 130; no inventories. 2025 budget?

### ▼ See the solution

The official solution is not published on Moodle yet: this one applies the course method, step by step.

| 2025 budget (k€) | Calculation | Amount |
|---|---|---|
| Equipment sales | 6,000 × 1.10 × 1.06 | 6,996 |
| Maintenance contracts | 1,800 × 1.06 | 1,908 |
| **Revenue** | | **8,904** |
| Cost of sales | 4,600 × 1.06 × 1.03 | -5,022.3 |
| **Gross margin** | | **3,881.7** |
| Marketing | 600 + 250 | -850 |
| Distribution | 150 × 1.06 | -159 |
| Client maintenance | 1,000 + 130 | -1,130 |
| Administration | unchanged | -900 |
| **Operating profit** | | **842.7** |

**Smell Good (launch budget), part I** Moodle statement; official Smell Good solution not open yet **Proposed solution**

Perfume sold at 250 excl. tax, production cost 90; base forecast 10,000 units. Samples: 20,000 for 20,000, 150 extra sales per 100 samples. Media: 300,000 for +15,000 units. Distributor discount: 5% up to 14M of sales. Part I - build the launch income statement using this exact format: Revenues - trade conditions - cost of sales - samples - media = profit.

### ▼ See the solution

The official solution is not published on Moodle yet: this one applies the course method, step by step.

Volume = 10,000 + 20,000 × 1.5 + 15,000 = **55,000 units**.

| | **€** |
|---|---|
| Revenue (55,000 × 250) | 13,750,000 |
| Distributor discount 5% | -687,500 |
| Cost of sales (55,000 × 90) | -4,950,000 |
| Samples | -20,000 |
| Media | -300,000 |
| **Profit** | **7,792,500** |

Part II (2% market share, about 110,000 units): team 1 can add 30,000 samples (+35,000 units, 25,000), 50 TV spots (+10,000, 300,000) and 100 magazine pages (+10,000, 130,000); team 2 compares the price/discount combinations from the table (e.g. 240 with a 9% discount gives 26.4M, i.e. 110,000 units). Rebuild the income statement of each scenario the same way.

**Smell Good, part II (market-share target)** Moodle statement Open exercise

Last-minute emergency: within the Part I scenario, the launch reaches only a 1% market share, while Fragrance & Luxury always targets 2% for a new product. Propose a new P&L that meets the 2% target, choosing one of two viewpoints. **Team 1 (advertising and samples):** extra TV spots (50 spots 300,000 → +10,000 units; 75 spots 400,000 → +13,000; 100 spots 475,000 → +15,000), extra magazine pages (30 pages 50,000 → +5,000; 60 pages 100,000 → +8,000; 100 pages 130,000 → +10,000), extra samples (20,000 samples 20,000 → +30,000 units; 25,000 samples 22,500 → +33,000; 30,000 samples 25,000 → +35,000). **Team 2 (price and discount levers):** gross revenue forecasts before trade discount - at price 250: 13,750,000 (5% discount) or 22,500,000 (9%); at 240: 23,000,000 (5%) or 26,400,000 (9%); at 225: 24,750,000 (5%) or 25,875,000 (9%).

▼ See the approach

Open discussion exercise with no single numeric answer: pick a lever set, recompute volumes, revenues, trade discount and costs in the Part I format, and check the profit against the 2% market-share target. Compare the cost of extra volume under each lever (advertising/samples vs price cut vs deeper discount) and defend the trade-off.

**Puyricard (margins, full costs and decisions)** Official Puyricard solution not open on Moodle yet **Proposed solution**

Chocolatier: chocolates (55/kg, 70,000 kg), candied chestnuts (70/kg, 10,000 kg), fruit pastes (35/kg, 40,000 kg). Materials 14, 7 and 3/kg; labor 1 h, 2 h and 0.5 h per kg at 16/h; other fixed direct 30,000, 15,000 and 10,000. IT 300,000 (40% workshop, 60% general services); workshop 584,000 charged per k€ of labor; general services 1,486,000 charged per k€ of sales. (1) Margins and full costs. (2) Indicators. (3) Options: A drop the chestnuts (workshop indirects down to 500,000); B new machine (materials 6/kg, -10% time, depreciation 10,000); C 3,000 extra kg sold at -30% through works councils. Analyze each option and their possible combinations, from a financial and a strategic point of view: which proposition(s) would you suggest the company adopt, and why?

### ▼ See the solution

The official solution is not published on Moodle yet: this one applies the course method, step by step.

| | Chocolates | Chestnuts | Fruit pastes |
|---|---|---|---|
| Sales | 3,850,000 | 700,000 | 1,400,000 |
| Materials | -980,000 | -70,000 | -120,000 |
| Labor | -1,120,000 | -320,000 | -320,000 |
| **Contribution margin** | **1,750,000** | **310,000** | **960,000** |
| Other fixed direct | -30,000 | -15,000 | -10,000 |
| **Net margin (direct costs)** | **1,720,000** | **295,000** | **950,000** |
| Workshop (704,000 → 0.40 per € of labor) | -448,000 | -128,000 | -128,000 |
| General services (1,666,000 → 0.28 per € of sales) | -1,078,000 | -196,000 | -392,000 |
| **Profit** | **194,000** | **-29,000** | **430,000** |
| Full cost per kg | 52.23 | 72.90 | 24.25 |
| Contribution / sales | 45.5% | 44.3% | 68.6% |
| Profit / sales | 5.0% | -4.1% | 30.7% |

- **Option A**: we lose the chestnuts' net margin (295,000) and save 84,000 of workshop: profit -211,000 (595,000 → 384,000). The apparent chestnut deficit comes from allocated overheads, which stay. Reject.
- **Option B**: materials -10,000, labor -32,000 (0.2 h × 16 × 10,000 kg), depreciation +10,000 → **+32,000**.
- **Option C**: price 49/kg, variable cost 39/kg → +10/kg × 3,000 = **+30,000** (42,600 with machine B, variable cost 34.80). Risk to the premium image and cannibalization.
- Recommendation: **B**, and C if the brand image allows (B + C: about +74,600).

## § Key terms

| Term | Definition |
|---|---|
| **Asset** | Present economic resource controlled as a result of past events. |
| **Liability** | Present obligation to transfer an economic resource as a result of past events. |
| **Shareholders' equity** | Residual interest in assets after deducting liabilities. |
| **Revenue / expense** | Equity-changing flows, excluding contributions and distributions. |
| **Accruals** | Revenue and expenses are recognized when earned or incurred, not when cash moves. |
| **Matching** | Expenses are tied to the revenue they helped produce. |
| **FIFO / LIFO / WAC** | Inventory cost-flow assumptions: first-in-first-out, last-in-first-out (US GAAP only), weighted average cost. |
| **Depreciation / amortization / depletion** | Systematic consumption of PP&E, intangibles and natural resources. |
| **Net book value** | Cost - accumulated depreciation - impairment. |
| **Impairment** | Carrying amount above recoverable amount = max(fair value - costs of disposal, value in use). |
| **Provision** | Liability of uncertain timing or amount: present obligation, probable outflow, reliable estimate. |
| **Prepaid / accrued expense** | Paid before it relates (asset) / relates before it is paid (liability). |
| **Deferred / accrued revenue** | Collected before it is earned (liability) / earned before it is collected (asset). |
| **Contribution margin** | Price - variable cost per unit; breakeven = fixed costs / unit margin. |
| **Relevant cost** | Future cost that differs between options; sunk costs are never relevant. |
| **Cost pool / allocation base** | Grouping of indirect costs / measurable driver used to allocate them. |
| **Job vs process costing** | Cost per order or batch / average cost of identical units. |
| **ABC** | Activity-Based Costing: homogeneous activity pools charged by driver consumption. |
| **Flexible budget** | Budget recomputed at the actual volume with budgeted prices and unit costs. |

# § MCQ annex, with answers

## Class #1 · Overview of financial accounting (9 Oct.)

**[Class #1] Only events with a monetary impact (a potential effect on cash) are recorded in financial accounting. True or false?**

a. False, every important event is recorded
b. False, only cash flows are recorded
c. False, only sales are recorded
d. True

**Answer: True**

Monetary unit convention: a decision or a hire with no measurable transaction is not recorded (Compofruit case).

**[Class #1] Financial accounting is completely objective and leaves no room for judgment. True or false?**

a. True under IFRS only
b. True, except for SMEs
c. False, it involves decisions and subjectivity, hence rules and principles
d. True, it follows strict rules

**Answer: False, it involves decisions and subjectivity, hence rules and principles**

The slides stress it: accounting is not mechanical.

**[Class #1] Taking out a long-term bank loan affects:**

a. Retained earnings
b. Equity
c. Operating liabilities (suppliers)
d. Financial liabilities

**Answer: Financial liabilities**

Cash +X, loan (financial liability) +X; no effect on equity.

**[Class #1] Which is usually NOT a current asset?**

a. Cash
b. Equipment
c. Trade receivables
d. Inventory

**Answer: Equipment**

Equipment is used over several periods: a non-current asset.

**A company buys equipment for 100 on credit. Effect on the accounting equation?**

a. Assets +100, equity +100
b. Liabilities +100, equity −100
c. Assets unchanged
d. Assets +100, liabilities +100

**Answer: Assets +100, liabilities +100**

An asset comes in, a supplier liability appears; net income is untouched.

**A company collects 42 from a customer receivable. Effect on total assets?**

a. None: cash +42, receivables −42
b. +42
c. −42
d. +42 and net income +42

**Answer: None: cash +42, receivables −42**

A simple reclassification within assets (Busoni, transaction 6).

**Busoni sells on credit for 40 half of the goods bought for 60. Effect on net income?**

a. +20
b. −30
c. +40
d. +10

**Answer: +10**

Revenue 40, then cost of goods sold 60/2 = 30: net income +10.

**Verdi repays 12 on its bank loan. Effect on net income?**

a. −12
b. None
c. +12
d. −12 in finance costs

**Answer: None**

Repaying principal reduces cash and the liability, not equity. Only interest is an expense.

**Verdi: contribution of 150, loan of 60, equipment bought for 125 in cash. Total assets after these three transactions?**

a. 335
b. 275
c. 210
d. 85

**Answer: 210**

Cash 150 + 60 − 125 = 85, equipment 125: total 210 = liabilities 60 + capital 150.

**In the Verdi case, net income is 60 but operating cash flows are −1. Why?**

a. Interest is operating
b. 70 of sales are not collected and part of the expenses are not paid: net income ≠ cash
c. Equipment is an expense
d. Calculation error

**Answer: 70 of sales are not collected and part of the expenses are not paid: net income ≠ cash**

Accrual principle: revenues and expenses do not coincide with receipts and payments.

**Which measurement basis is easy to determine and verifiable, but does not track market value?**

a. Fair value
b. Value in use
c. Liquidation value
d. Historical cost

**Answer: Historical cost**

The advantage and limit of historical cost, per the slides.

**From 2027, which standard replaces IAS 1 for the presentation of financial statements?**

a. IAS 7
b. IFRS 9
c. IFRS 18
d. IFRS 15

**Answer: IFRS 18**

IFRS 18, adopted by the EU in February 2026, creates the operating profit and profit before financing and tax subtotals, among others.

## Class #2 · Operating cycle and current assets (16 Oct.)

**[Class #2] Under IFRS 15, which is NOT a condition for recognising revenue from a sale of goods?**

a. The contract is identified
b. Control of the goods is transferred to the customer
c. The cash for the sale has been collected
d. The amount of consideration can be determined

**Answer: The cash for the sale has been collected**

Accounting is accrual-based: collection is not required.

**[Class #2] When purchase costs rise, net income under FIFO is:**

a. Lower than under LIFO or weighted average
b. Sometimes higher, sometimes lower
c. Higher than under LIFO or weighted average
d. Identical

**Answer: Higher than under LIFO or weighted average**

FIFO releases the oldest, cheapest inventory: lower cost of goods sold.

**[Class #2] Why can year-end inventory be worth less than its carrying amount?**

a. Falling replacement prices
b. Physical deterioration
c. Obsolescence
d. All of these reasons

**Answer: All of these reasons**

Inventory is measured at the lower of cost and net realisable value.

**[Class #2] Which statement is false?**

a. Uncollectible receivables must be written off
b. A receivable is uncollectible when the debtor is insolvent
c. Writing off an uncollectible receivable creates an expense
d. An impairment on a doubtful receivable can never be reversed

**Answer: An impairment on a doubtful receivable can never be reversed**

An impairment is reversed if the risk decreases.

**Opening inventory 50 units at €100, purchase of 75 units at €110. Weighted average cost?**

a. 110
b. 105
c. 106
d. 108

**Answer: 106**

(5,000 + 8,250)/125 = 106 (Cameras case).

**Cameras case: closing inventory value under FIFO?**

a. 5,000
b. 5,495
c. 5,750
d. 6,890

**Answer: 5,750**

The 25 units bought at 110 and the 25 at 120 remain: 2,750 + 3,000 = 5,750.

**Mozart case: purchases of 200 units at €22 then 180 at €18; 260 units sold. Cost of goods sold under FIFO?**

a. 5,000
b. 5,480
c. 5,120
d. 5,227

**Answer: 5,480**

200 × 22 + 60 × 18 = 4,400 + 1,080 = 5,480.

**Mozart case (purchase prices falling: 22 then 18). Which method gives the highest net income?**

a. LIFO
b. Weighted average
c. FIFO
d. They all give the same

**Answer: LIFO**

Falling prices: LIFO releases the €18 units, the lowest cost of goods sold (5,000).

**Purchase price 1,000, transport 60, customs duties 40, trade discount 50. Acquisition cost of the inventory?**

a. 1,100
b. 1,150
c. 1,050
d. 1,000

**Answer: 1,050**

1,000 + 60 + 40 − 50 = 1,050: discounts are deducted, transport and duties are added.

**A receivable of 100 of which only 60 is expected to be collected. Effect on the year's net income?**

a. −60
b. None until the customer is liquidated
c. −40
d. −100

**Answer: −40**

Impairment of 40 through an expense (Bruckner example, year X1).

**Bruckner: accumulated impairment of 70 on a receivable of 100; the liquidated customer pays 10. Expense in the liquidation year?**

a. −70
b. −90
c. −20
d. 0

**Answer: −20**

Total loss 90, of which 70 already recognised: net expense 20.

**A school collects in December X1 the tuition fees for the first term of X2. In X1 it must record:**

a. Accrued revenue
b. Deferred revenue (a liability)
c. Revenue of X1
d. A provision

**Answer: Deferred revenue (a liability)**

Schultz exercise (a): the revenue belongs to X2.

**Consignment sales: when does the consignor recognise revenue?**

a. When the contract is signed
b. On delivery to the consignee
c. At year-end
d. When the consignee sells the goods

**Answer: When the consignee sells the goods**

Schultz exercise (d): unsold goods still belong to the consignor.

## Class #3 · Investing cycle and non-current assets (23 Oct.)

**[Class #3] When an asset is sold mid-year, depreciation is recorded:**

a. For half the year
b. For the full year
c. From the last depreciation charge (end of the previous period) to the disposal date
d. Not at all

**Answer: From the last depreciation charge (end of the previous period) to the disposal date**

Pro rata temporis up to the disposal.

**[Class #3] What happens when the carrying amount reaches the residual value?**

a. It is revalued
b. The asset disappears from the balance sheet
c. The asset must be sold
d. The asset can stay in use, on the balance sheet at its residual value

**Answer: The asset can stay in use, on the balance sheet at its residual value**

Depreciation stops; the asset stays if it is still useful.

**[Class #3] Amortising an intangible over the period it provides benefits applies the principle of:**

a. Going concern
b. Matching
c. Accruals
d. Prudence

**Answer: Matching**

The consumption of the asset is matched to the revenues it generates.

**[Class #3] Brands can always be recognised as intangible assets. True or false?**

a. True if it is well known
b. False: an internally generated brand is not recognised
c. True under US GAAP only
d. True

**Answer: False: an internally generated brand is not recognised**

Only acquired brands are recognised.

**Cost 11,000, residual value 1,000, 5 years, straight-line. Annual charge?**

a. 2,000
b. 1,800
c. 2,750
d. 2,200

**Answer: 2,000**

(11,000 − 1,000) × 20% = 2,000.

**A 15,000 machine over 10 years, double declining balance. Year 2 charge?**

a. 2,700
b. 1,500
c. 3,000
d. 2,400

**Answer: 2,400**

20% rate: 3,000 in year 1, then 20% × 12,000 = 2,400.

**Same machine (15,000, double declining over 10 years): carrying amount after 2 years?**

a. 10,200
b. 9,000
c. 12,000
d. 9,600

**Answer: 9,600**

15,000 − 3,000 − 2,400 = 9,600.

**Price 100, customs duties 5, transport 15, engineers' fees 30. Entry cost of the equipment?**

a. 105
b. 100
c. 150
d. 120

**Answer: 150**

All directly attributable costs to bring the asset to working condition are included.

**Equipment with a carrying amount of 40 sold for 45. Effect on net income?**

a. +45
b. +5
c. None
d. −5

**Answer: +5**

Gain on disposal = price − carrying amount = 5, presented net (IAS 16.68).

**Which is NOT depreciated?**

a. Software
b. A patent
c. A building
d. Land

**Answer: Land**

Land is not consumed: no depreciation.

**Which can be recognised as an asset?**

a. Goodwill acquired in a takeover
b. Training costs
c. Internally generated goodwill
d. Research expenditure

**Answer: Goodwill acquired in a takeover**

Only acquired goodwill (price − fair value of identifiable net assets) is recognised.

**Amorti case: launch advertising of €100,000. Treatment?**

a. Expense of the year
b. Asset amortised over 5 years
c. Provision
d. Asset not amortised

**Answer: Expense of the year**

Advertising is never an asset: immediate expense.

**Amorti case: year 1 net income under straight-line (sales 1,000,000, other expenses 800,000, licence 15,000, advertising 100,000, line 60,000)?**

a. 200,000
b. −35,000
c. 125,000
d. 25,000

**Answer: 25,000**

1,000,000 − 800,000 − 15,000 − 100,000 − 60,000 = 25,000.

**Amorti case: over the 5 years, total net income under straight-line and declining balance is:**

a. Impossible to compare
b. Higher under declining balance
c. Identical
d. Higher under straight-line

**Answer: Identical**

The method only changes the timing of the allocation.

## Class #4 · Financing cycle, equity and liabilities (30 Oct.)

**ACME: share capital €5,000k, par value €25. Number of shares?**

a. 50,000
b. 2,000
c. 250,000
d. 200,000

**Answer: 200,000**

5,000,000 / 25 = 200,000.

**ACME raises €5,000k issuing shares at €100 (par value €25). Share premium?**

a. €5,000k
b. €3,750k
c. 0
d. €1,250k

**Answer: €3,750k**

50,000 shares × (100 − 25) = €3,750k; capital increases by €1,250k.

**ACME after the increase (equity €12,300k, 250,000 shares). Book value per share?**

a. €100
b. €25
c. €110
d. €49.2

**Answer: €49.2**

12,300,000 / 250,000 = €49.2: different from par value and market price.

**Gerry Co.: opening reserves 50, net income 70, dividends declared 30. Reserves after appropriation?**

a. 70
b. 120
c. 90
d. 50

**Answer: 90**

50 + 70 − 30 = 90; the 30 become a liability (dividends payable).

**Dividends declared but not yet paid appear:**

a. In expenses
b. In equity
c. In liabilities
d. Nowhere

**Answer: In liabilities**

They become an obligation to shareholders.

**Which difference between liabilities and equity is correct?**

a. Lenders control management
b. Interest is due whatever the result
c. Dividends are generally tax-deductible
d. Equity must be repaid

**Answer: Interest is due whatever the result**

Interest is due even in a loss year, dividends are not.

**Par value €10, 1,000 shares; issue of 160 shares at €25. Share capital after the issue?**

a. 14,000
b. 12,400
c. 10,000
d. 11,600

**Answer: 11,600**

1,160 × 10 = 11,600 (Q7 of the sample paper).

## Class #5 · End-of-Year adjustments (6 Nov.)

**[Class #5] Loan to an employee, annual interest of 120 due every 31 March. Interest revenue at 31 December?**

a. 0
b. 30
c. 90
d. 60

**Answer: 90**

Nine months accrued (April to December): 120 × 9/12 = 90, as accrued revenue.

**[Class #5] Which is an adjusting entry?**

a. Paying salaries already accrued
b. Returning defective inventory
c. Recording accrued interest payable
d. Collecting a receivable

**Answer: Recording accrued interest payable**

Accrued interest not yet due is recorded at closing.

**[Class #5] Annual rent of 1,200 paid in February N for 1 February N to 31 January N+1. Entry at end of N?**

a. Accrued expense
b. Prepaid expense
c. Deferred revenue
d. Accrued revenue

**Answer: Prepaid expense**

The month of January N+1 (100) was paid in advance.

**[Class #5] Electricity billed on 31 January N for the last two months. No entry is needed at end of N−1. True or false?**

a. True if the bill is small
b. False: an accrued expense must be recorded for December
c. False: a provision is needed
d. True

**Answer: False: an accrued expense must be recorded for December**

December consumption belongs to N−1.

**[Class #5] Warranty costs are expensed:**

a. When the warranted products are sold
b. Never
c. When the repairs take place
d. At the end of the warranty

**Answer: When the warranted products are sold**

A provision is made at the time of sale to match the expense to the revenue.

**[Class #5] A contingent liability is recognised:**

a. In revenue
b. It is not recognised, only disclosed in the notes
c. As an asset
d. As a liability

**Answer: It is not recognised, only disclosed in the notes**

The conditions for a provision are not met.

**[Class #5] A restructuring provision can be made as soon as management has decided to close an activity. True or false?**

a. True
b. False: a formal plan and implementation started or an announcement are required
c. True if the amount is known
d. True under IFRS

**Answer: False: a formal plan and implementation started or an announcement are required**

Case of the closure decided on 12 December without announcement: no provision.

**Rent of 9,000 paid on 1 November N for November, December and January. Rent expense for year N?**

a. 3,000
b. 9,000
c. 6,000
d. 0

**Answer: 6,000**

Three months paid, two used in N: 6,000; 3,000 as a prepaid expense.

**Annual subscription of 1,200 collected on 1 July N. Deferred revenue at 31 December N?**

a. 1,200
b. 0
c. 600
d. 300

**Answer: 600**

Six months remain to be served in N+1.

**Invoice of 900 issued on 1 February N+1 for November, December and January. Accrued revenue at 31 December N?**

a. 300
b. 0
c. 900
d. 600

**Answer: 600**

Two of the three months belong to N.

**10,000 warranted products: 75% no defect, 20% small defect (cost 1), 5% major defect (cost 10). Provision?**

a. 2,000
b. 5,000
c. 10,000
d. 7,000

**Answer: 7,000**

10,000 × 20% × 1 + 10,000 × 5% × 10 = 2,000 + 5,000.

**Carrying amount 1,200; fair value less costs of disposal 950; value in use 800. Impairment loss?**

a. 400
b. 250
c. 150
d. 0

**Answer: 250**

Recoverable amount = max(950; 800) = 950; loss 1,200 − 950.

**Imp case: carrying amount 600,000; value in use 800,000; net selling price 950,000. What to do?**

a. Nothing: the recoverable amount exceeds the carrying amount
b. Impair by 350,000
c. Impair by 200,000
d. Revalue to 950,000

**Answer: Nothing: the recoverable amount exceeds the carrying amount**

No impairment, and no upward revaluation under historical cost.

**Can an impairment loss on goodwill be reversed?**

a. Yes if the value recovers
b. Yes, always
c. Yes under US GAAP only
d. No, never

**Answer: No, never**

IAS 36 prohibits reversal for goodwill.

**Lawsuit: provision of 9M on the balance sheet, final settlement of 9.5M. Effect on the settlement year's net income?**

a. −9.5M
b. 0
c. −0.5M
d. +9M

**Answer: −0.5M**

The provision covers 9M; the additional expense is 0.5M.

**Flossie Flowers: X2 loan interest (8,362) not recorded at 31/12/X2. Entry?**

a. Provision of 26,380
b. Nothing until payment
c. Prepaid expense
d. Accrued expense of 8,362

**Answer: Accrued expense of 8,362**

The X2 interest is incurred: accrued expense.

**Offenbach: six-month insurance paid on 1 September X1. At 31 December X1:**

a. Accrued expense
b. Deferred revenue
c. Prepaid expense of 2 months
d. Nothing

**Answer: Prepaid expense of 2 months**

January and February X2 were paid in advance.

**Offenbach: one year's rent collected on 1 April X1 for a leased office. At 31 December X1:**

a. Nothing
b. Prepaid expense
c. Deferred revenue of 3 months
d. Accrued revenue

**Answer: Deferred revenue of 3 months**

Three months of rent belong to X2.

**An airline must overhaul its planes every three years. A provision for future overhauls?**

a. No: no present obligation arising from a past event
b. Yes, one third each year
c. Yes, it is a legal obligation
d. Yes if the amount is reliable

**Answer: No: no present obligation arising from a past event**

Provisor case (d): the overhaul depends on future operations.

**A lawsuit whose loss is probable and estimated at €350,000:**

a. Nothing
b. Accrued expense
c. Provision of €350,000
d. Contingent liability in the notes

**Answer: Provision of €350,000**

All three conditions for a provision are met (Provisor c).

## Class #6 · Statement of cash flows (13 Nov.)

**[Class #6] Debt still owed at the end of year N never appears in N's cash flow statement. True or false?**

a. True except for loans
b. True: the statement only shows flows
c. False
d. False under the indirect method

**Answer: True: the statement only shows flows**

Balances are on the balance sheet; the statement only presents receipts and payments.

**[Class #6] Under the direct method, which does NOT appear in operating cash flows?**

a. Net income
b. Payments to suppliers
c. Collections from customers
d. Payments to employees

**Answer: Net income**

Net income is the starting point of the indirect method.

**[Class #6] Which is NOT a non-cash item?**

a. Depreciation charge on tangible fixed assets
b. Payment of tax
c. Amortisation charge on intangibles
d. Reversal of a provision

**Answer: Payment of tax**

Paying tax is a real cash outflow.

**[Class #6] Which increases cash?**

a. Cash sales
b. All of these
c. Dividends received
d. Collection of receivables

**Answer: All of these**

Each one is a cash inflow.

**Net income 100; depreciation 50; trade receivables +30; trade payables +20. Operating cash flows (indirect method)?**

a. 140
b. 160
c. 200
d. 100

**Answer: 140**

100 + 50 − 30 + 20.

**The purchase of a machine paid in cash is reported in:**

a. Operating cash flows
b. Investing cash flows
c. Change in working capital
d. Financing cash flows

**Answer: Investing cash flows**

Acquisition of a fixed asset.

**From 2027 (IAS 7 amended by IFRS 18), where does interest paid go for most companies?**

a. Investing cash flows
b. Free choice
c. Financing cash flows
d. Operating cash flows

**Answer: Financing cash flows**

Interest and dividends received in investing, paid in financing.

**Beethoven case: operating cash flows of X2?**

a. −60
b. 0
c. +260
d. +120

**Answer: −60**

1,620 − 570 − 240 − 450 − 300 − 75 − 45 = −60.

**Beethoven case: net income of X2?**

a. 200
b. 80
c. 120
d. 360

**Answer: 120**

Pre-tax income 200, tax 40% = 80, net income 120.

**A capital increase of 240 paid in cash is reported in:**

a. Not reported
b. Operating cash flows
c. Investing cash flows
d. Financing cash flows

**Answer: Financing cash flows**

Share issue: financing.

## Class #7 · Introduction to Management Accounting; costing terms and purpose (20 Nov.)

**Price £70, variable cost £42, fixed costs £84,000. Break-even point?**

a. 3,000 units
b. 1,200 units
c. 2,500 units
d. 2,000 units

**Answer: 3,000 units**

84,000 / (70 − 42) = 3,000 (Dresses by Mary).

**Same case: contribution margin ratio?**

a. 60%
b. 28%
c. 40%
d. 70%

**Answer: 40%**

28 / 70 = 40%.

**Montvert: price €2, variable cost €1.20, fixed costs €3.6M, 10M units. Net income?**

a. €12.4M
b. €4.4M
c. €8M
d. €3.6M

**Answer: €4.4M**

10M × 0.80 − 3.6M.

**Montvert: break-even point in revenue?**

a. €4.5M
b. €3.6M
c. €9M
d. €7.2M

**Answer: €9M**

4.5M units × €2 (or 3.6M / 40%).

**Montvert: variable cost rises by €0.16. New break-even point?**

a. 5.625M units
b. 5M units
c. 6M units
d. 4.5M units

**Answer: 5.625M units**

3.6M / 0.64.

**Walk-About: price £30, variable cost £21, fixed costs £360,000. Net income at 35,000 pairs?**

a. −£45,000
b. +£45,000
c. +£315,000
d. 0

**Answer: −£45,000**

35,000 × 9 − 360,000.

**Walk-About: indifference volume between the commission plan (margin 9, fixed 360,000) and the fixed plan (margin 10.50, fixed 441,000)?**

a. 42,000
b. 58,000
c. 40,000
d. 54,000

**Answer: 54,000**

1.5 Q = 81,000.

**Speed Truck: price €1,200, variable cost €250, fixed costs €104,000. Trucks for a profit of €57,500?**

a. 135
b. 160
c. 110
d. 170

**Answer: 170**

(57,500 + 104,000) / 950.

**Noel: variable cost €8.50, full cost €13.50, spare capacity. Special order of 5,000 towels at €11.50?**

a. Reject: price below full cost
b. Reject: −€10,000
c. Indifferent
d. Accept: +€15,000

**Answer: Accept: +€15,000**

Only variable costs are relevant: (11.50 − 8.50) × 5,000.

**Tiilikainen at full capacity: motorbike engine (margin €240, 2 h) and boat engine (€375, 5 h). Which engine to favour?**

a. Boat: higher margin ratio
b. Boat: higher unit margin
c. Both equally
d. Motorbike: €120 per hour vs 75

**Answer: Motorbike: €120 per hour vs 75**

With a scarce resource, rank by margin per unit of resource.

**Kruger: fixed €94,500, variable €52 per bike. Unit cost for 600 bikes?**

a. €157.50
b. €209.50
c. €52
d. €146.50

**Answer: €209.50**

(94,500 + 31,200) / 600: the €146.50 unit cost computed at 1,000 bikes would understate it.

**Which cost is NOT relevant for a decision?**

a. An opportunity cost
b. A differential revenue
c. A future cost that differs between options
d. A sunk cost already incurred

**Answer: A sunk cost already incurred**

Sunk costs do not change with the decision.

**My new car: differential cost per business kilometre?**

a. €0.286
b. €0.126
c. €0.358
d. €0.35

**Answer: €0.286**

€4,290 of extra costs for 15,000 km; the €0.35 offer is acceptable.

**Kuniyakino: the sales commission of the Management department head is:**

a. Indirect and fixed
b. Direct and fixed
c. Direct and variable
d. Indirect and variable

**Answer: Direct and variable**

Specific to the department and linked to sales.

**Kuniyakino: the store's single electricity bill is, for the Management department:**

a. Indirect and variable
b. Indirect and fixed
c. Direct and fixed
d. Direct and variable

**Answer: Indirect and fixed**

It covers the whole store and does not depend on sales.

## Class #8 · Cost allocation: job order costing and process-costing (27 Nov.)

**Immoservices: £21,000 of indirect costs, 200 hours. Rate per hour?**

a. £210
b. £100
c. £105
d. £70

**Answer: £105**

21,000 / 200.

**Immoservices (single cost centre): cost of the Pemberley engagement (104 h at £70 + indirect costs)?**

a. £16,800
b. £18,200
c. £23,660
d. £7,280

**Answer: £18,200**

7,280 + 104 × 105 = 18,200; price £23,660 with 30%.

**Immoservices with two centres (administration £35/h, documentation £1,400 per request; Pemberley 2 requests). Cost of Pemberley?**

a. £21,280
b. £18,200
c. £13,720
d. £10,920

**Answer: £13,720**

7,280 + 3,640 + 2,800.

**Slides example, direct method: maintenance 600,000 allocated on the main centres' hours (machining 2,400 h, assembly 4,000 h). Machining's share?**

a. 180,000
b. 225,000
c. 375,000
d. 240,000

**Answer: 225,000**

600,000 × 2,400 / 6,400.

**Same example, sequential method: maintenance allocated first (20% to IT). IT cost to be allocated next?**

a. 236,000
b. 116,000
c. 120,000
d. 600,000

**Answer: 236,000**

116,000 + 120,000.

**WattWatt: overhead 2,400,000, direct labour 1,500,000. Overhead charged to contract Alpha (labour 300,000)?**

a. 400,000
b. 300,000
c. 480,000
d. 562,500

**Answer: 480,000**

Rate 1.6 × 300,000.

**Which industry is typical of process costing?**

a. Oil refining
b. Building a house
c. A consulting engagement
d. An advertising campaign

**Answer: Oil refining**

Mass production of identical units.

**Alp'Ski: total of the Cutting centre after allocating support centres?**

a. 124,000
b. 91,000
c. 75,000
d. 35,000

**Answer: 91,000**

35,000 + 10,000 (maintenance) + 30,000 (IT) + 16,000 (purchasing).

**Should a product's full cost be used to decide to drop that product?**

a. Yes, that is its purpose
b. No: a differential cost and revenue analysis is needed
c. Yes under ABC
d. Yes if the product is loss-making

**Answer: No: a differential cost and revenue analysis is needed**

Allocated indirect costs often remain after dropping it (cf. Puyricard, option A).

## Class #9 · Activity-Based Costing (4 Dec.)

**McCarthy: cutting 231,000 €; 3,600 h (mass retail) and 250 h (institutional catering). Cost per cutting hour?**

a. €231
b. €60
c. €64
d. €924

**Answer: €60**

231,000 / 3,850.

**McCarthy under ABC: cost of a kg of fries for institutional catering?**

a. €1.025
b. €1.133
c. €0.62
d. €1.19

**Answer: €0.62**

0.15 + 0.08 + 0.12 + 0.15 + 0.12.

**Stella Cuero: machine setups €24,250 for 2,500 setups. Cost per setup?**

a. €9.70
b. €3.75
c. €5.00
d. €10.00

**Answer: €9.70**

24,250 / 2,500.

**Stella Cuero: Toledo's result under ABC (sales 240,000, direct 218,400, ABC indirect 15,690)?**

a. +5,910
b. +21,600
c. −3,000
d. −15,690

**Answer: +5,910**

240,000 − 218,400 − 15,690; under the cost-centre method, Toledo lost 3,000.

**Alp'Ski under ABC: which ski becomes loss-making?**

a. EXTREME
b. Both
c. None
d. GRIMPER

**Answer: GRIMPER**

GRIMPER (small batches, many setups and drying): −7,824.

**What is a limit of ABC?**

a. A heavy database to build and maintain
b. It distorts the cost of complex products
c. It only applies to manufacturing
d. It ignores indirect costs

**Answer: A heavy database to build and maintain**

Number of drivers × number of products.

**Under ABC, reducing the number of drivers a product consumes:**

a. Has no effect
b. Always reduces the company's total cost
c. Mostly shifts costs between products
d. Eliminates fixed costs

**Answer: Mostly shifts costs between products**

Stella Cuero correction, question 4.

## Class #10 · Introduction to budget and variance analysis (11 Dec.)

**A la Coque: flexible-budget operating income (15,000 units, price €40, variable €4.50, fixed 42,000)?**

a. 600,000
b. 490,500
c. 485,000
d. 384,000

**Answer: 490,500**

600,000 − 67,500 − 42,000.

**A la Coque: sales-volume variance on operating income?**

a. +106,500 favourable
b. +101,000 favourable
c. −5,500 unfavourable
d. +120,000 favourable

**Answer: +106,500 favourable**

Flexible budget 490,500 − static budget 384,000.

**A la Coque: flexible-budget variance on operating income?**

a. −21,000 unfavourable
b. −5,500 unfavourable
c. +5,500 favourable
d. +106,500 favourable

**Answer: −5,500 unfavourable**

Actual 485,000 − flexible 490,500.

**Castelo Branco: 2025 revenue (k€) with equipment 6,000 × 1.10 × 1.06 and contracts 1,800 × 1.06?**

a. 8,904
b. 9,200
c. 8,586
d. 7,800

**Answer: 8,904**

6,996 + 1,908.

**Smell Good: volume sold with 20,000 samples (150 sales per 100) and the media campaign (+15,000), base 10,000?**

a. 25,000
b. 40,000
c. 45,000
d. 55,000

**Answer: 55,000**

10,000 + 30,000 + 15,000.

**Smell Good: launch result (55,000 units at €250, 5% discount, cost €90, samples €20,000, media €300,000)?**

a. €8,800,000
b. €7,472,500
c. €8,480,000
d. €7,792,500

**Answer: €7,792,500**

13,750,000 − 687,500 − 4,950,000 − 20,000 − 300,000.

**Puyricard, option A (dropping the marrons glacés: net margin of 295,000 lost, 84,000 of savings):**

a. No effect
b. +29,000 on operating income
c. −211,000 on operating income
d. +84,000

**Answer: −211,000 on operating income**

The indirect costs allocated to the marrons mostly remain.

**Cityline: the Dualtron product manager proposes €500 for 16,000 units. Overall result:**

a. Increases by 164,706
b. Stays the same
c. Increases by 100,000
d. Decreases by 100,000

**Answer: Decreases by 100,000**

Dualtron's direct-cost margin falls by 100,000; overheads are only redistributed.

**Which statement about the budget is correct?**

a. It only contains financial items
b. It always covers five years
c. It is the quantified expression of a forward-looking vision over a limited period
d. It is mandatory and standardised

**Answer: It is the quantified expression of a forward-looking vision over a limited period**

Definition from the session 10 slides.

## Sample exam (exfa)

**[Sample paper FA, Q1] Cash-collected sales 2,000; purchases paid 1,000; depreciation 100; interest 50; receivables decrease 30; trade payables increase 60; non-operating interest. Operating cash flows?**

a. 1,090
b. 1,240
c. 1,010
d. 1,060

**Answer: 1,090**

2,000 − 1,000 + 30 + 60; depreciation is not a flow and interest is excluded.

**[Sample paper FA, Q2] An increase in trade payables has what effect on cash?**

a. None
b. Negative or positive
c. Positive
d. Negative

**Answer: Positive**

Unpaid purchases preserve cash.

**[Sample paper FA, Q3] Which is NOT an accounting principle or convention?**

a. Accruals
b. Incurring
c. Matching
d. Prudence

**Answer: Incurring**

"Incurring" is not a principle.

**[Sample paper FA, Q4] Which is recognised as an asset?**

a. Internally generated goodwill
b. Advertising expected to triple sales
c. A patent
d. Research expenditure

**Answer: A patent**

Only the patent is an identifiable asset.

**[Sample paper FA, Q5] Equity at end of 2015: 19,200 (including net income 1,200). In 2016: net income 600, dividends 500, land contribution of 800. Equity and net income at end of 2016?**

a. 20,100 and 100
b. 20,100 and 600
c. 19,300 and 1,400
d. 19,300 and 600

**Answer: 20,100 and 600**

19,200 + 600 − 500 + 800.

**[Sample paper FA, Q6] Equipment of 1,200; depreciation 150 then 175; impairment of 50. Carrying amount at end of 2016?**

a. 625
b. 825
c. 775
d. 875

**Answer: 825**

1,200 − 375.

**[Sample paper FA, Q7] 1,000 shares of €10 par value; contribution of a building worth €3,000 and €1,000 against shares issued at €25. Which statement is false?**

a. Capital is 10,000 after the transaction
b. Equity increases by 4,000
c. 160 shares are issued
d. The premium increases by 2,400

**Answer: Capital is 10,000 after the transaction**

Capital rises to 11,600.

**[Sample paper FA, Q8] Which situation cannot give rise to a provision?**

a. A restructuring
b. An accrued liability
c. A sale with a warranty
d. A lawsuit brought by a customer

**Answer: An accrued liability**

Timing and amount are almost certain: it is not a provision.

**[Sample paper FA, Q9] Goods purchased for 2,000; half resold for 1,800; next year's rent paid 300. Net income?**

a. 500
b. 300
c. 100
d. 800

**Answer: 800**

1,800 − 1,000; the prepaid rent is not an expense of the year.

**[Sample paper FA, Q10] A car dealer's purchase of cars belongs to which cycle?**

a. Operating
b. Financing
c. Investing
d. Non-recurring

**Answer: Operating**

For the dealer, cars are inventory.

**[Sample paper FA, Q11] A €10,000 machine, 5 years, estimated resale €1,000. Straight-line depreciation in year 1?**

a. 2,000
b. 1,800
c. 2,200
d. None of these

**Answer: 1,800**

(10,000 − 1,000) / 5.

**[Sample paper FA, Q12] Same machine under double declining balance: year 1 charge?**

a. 2,000
b. 3,000
c. 3,600
d. 4,000

**Answer: 4,000**

40% × 10,000 (residual value is not deducted under declining balance).

## Sample exam (exfa)

**Cityline: manufacturing overhead rate per direct-labour hour (€6M; 20,000 h + 60,000 h)?**

a. €100
b. €60
c. €300
d. €75

**Answer: €75**

6,000,000 / 80,000.

**Cityline, single allocation base: Dualtron's result?**

a. +650,000
b. +4,400,000
c. −100,000
d. +2,950,000

**Answer: −100,000**

9,000,000 − 4,600,000 − 4,500,000.

**Cityline, four centres: cost of the Machines centre after allocating support centres?**

a. 3,000,000
b. 2,000,000
c. 4,200,000
d. 1,800,000

**Answer: 4,200,000**

2,000,000 + 1,000,000 (plant) + 1,200,000 (quality).

**Cityline, four centres: Dualtron's result?**

a. +64,706
b. +650,000
c. +2,200,000
d. −100,000

**Answer: +650,000**

9,000,000 − 4,600,000 − 3,750,000.

**A la Coque: variable-cost variance (actual 75,000, static budget 54,000)?**

a. 21,000 favourable
b. 13,500 unfavourable
c. 7,500 unfavourable
d. 21,000 unfavourable

**Answer: 21,000 unfavourable**

Costs higher than planned: unfavourable.

## Sample exam (exfa)

**Speed Truck: annual fixed costs (salaries 43,200, depreciation 15,000, other 45,800)?**

a. 104,000
b. 58,800
c. 149,000
d. 89,000

**Answer: 104,000**

43,200 + 15,000 + 45,800.

**Speed Truck: break-even point (price €900, variable cost €250, fixed costs €104,000)?**

a. 170 trucks
b. 160 trucks
c. 180 trucks
d. 116 trucks

**Answer: 160 trucks**

104,000 / 650.

**Stella Cuero: cost per sewing hour (€60,750 for 11,250 machine-hours)?**

a. €4.67
b. €5.40
c. €2.50
d. €6.08

**Answer: €5.40**

60,750 / 11,250; cutting costs €4.67/h.

**Stella Cuero, cost-centre method: which bag is loss-making?**

a. Sevilla
b. Granada
c. None
d. Toledo

**Answer: Toledo**

Toledo: 240,000 − 243,000 = −3,000.